Math Comparative Advantage and the Gender Gap in STEM: From Upper-Secondary Tracks to University Degrees

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Working Paper: October 2026

Abstract

Women remain underrepresented in STEM, particularly in math-intensive fields. Using administrative records for 219,638 Catalan students, we examine how mathematics achievement, within-student comparative advantage, and peer rank account for gender gaps in upper-secondary Science-track and university STEM-degree enrollment. Comparative advantage, measured as the difference between standardized mathematics and reading scores, accounts for 48.5% of the 16.6-percentage-point conditional Science-track gap but 18.5% of the 29.2-point STEM-degree gap among university entrants. The latter share rises to 38.0% when STEM includes life sciences and medicine. Comparative advantage accounts for more than mathematics achievement primarily because girls and boys differ more in it; their contributions overlap substantially and are sensitive to measurement error. Peer rank adds little to the gap already accounted for by mathematics achievement. At university, girls' enrollment rises about half as steeply as boys' with comparative advantage on the percentage-point scale, and the gap exceeds 30 points among the strongest mathematics students. Measures taken in 6th grade account for substantially less of both gaps. Academic profiles thus account for a substantial share of the Science-track gap but leave most of the university gap unaccounted for.

Keywords: Gender gap; STEM; comparative advantage; peer rank; track choice; field of study; mathematics achievement.

JEL Codes: J16, J24, I21, I23, I24.

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1 Introduction

Women have made substantial gains in educational attainment (Becker et al., 2010; Klasing and Milionis, 2020), yet remain markedly underrepresented in science, technology, engineering, and mathematics (STEM). Women account for only about 35% of STEM graduates worldwide, a share that has remained virtually unchanged over the past decade, with particularly pronounced underrepresentation in engineering and information and communication technology (ICT) (UNESCO, 2021; Global Education Monitoring Report Team, 2024). Yet average gender differences in mathematics achievement at school age are small in most education systems (Breda and Napp, 2019; Borgonovi et al., 2021). The persistence of large gender differences in STEM specialization alongside converging educational attainment and small average gaps in mathematics performance suggests that absolute mathematics achievement alone is unlikely to account for the gender gap in STEM choices. We study whether the same mathematics achievement is associated with different patterns of specialization depending on how it compares with other salient dimensions of students' academic performance.

We focus on two observable comparison dimensions emphasized in the literature on academic self-evaluation (Möller et al., 2009): a student's own performance in other subjects and the mathematics performance of their peers. The first is an intra-individual comparison across subjects, the educational analogue of comparative advantage in Roy-type models of self-selection (Roy, 1951; Willis and Rosen, 1979); the second captures peer-relative standing within mathematics. The two need not coincide. A student may rank highly in mathematics among their peers while performing even better in reading, whereas another may have mathematics as their strongest subject despite occupying a more moderate position in the mathematics distribution. We do not observe students' subjective perceptions of their performance. Rather, we study observable academic profiles corresponding to these two comparison dimensions and ask how much they account for gender differences in STEM specialization beyond absolute mathematics achievement. More specifically, we ask how their accounting contribution changes across two successive realized specialization decisions and how it differs when performance is measured at different stages of schooling.

Evidence that cross-subject academic profiles matter for educational specialization predates the recent debate on gender gaps in STEM. Jonsson (1999) shows that students' relative strengths across quantitative and non-quantitative subjects are associated with upper-secondary program choice in Sweden, while Dekhtyar et al. (2018) document that numerical strengths relative to verbal strengths predict the quantitative orientation of subsequent education and occupations. Breda and Napp (2019) bring this mechanism to the center of the STEM gender-gap debate. Using PISA 2012 data from 64 countries, they show that girls' performance is systematically tilted toward reading relative to boys' and that this differential accounts for a large share of the gender gap in forced-choice intentions to pursue mathematics-intensive studies and careers. Subsequent evidence yields substantially smaller and more heterogeneous accounting

contributions. Wan et al. (2023), using PISA 2018 and the US High School Longitudinal Study, find much smaller shares for open-ended career expectations and for both intended and realized college fields. Studies of realized applications and enrollment likewise show that cross-subject academic profiles are associated with specialization but leave substantial gender differences unexplained (Riegle-Crumb et al., 2012; Delaney and Devereux, 2019; Dekhtyar et al., 2018; Aucejo and James, 2021). Importantly, these studies do not all measure the same object: some construct an explicit within-student differential (Breda and Napp, 2019; Sadowski and Zawistowska, 2020), whereas others enter mathematics and verbal skills separately.

A related literature examines students' performance relative to their peers. Subject-specific rank is associated with later educational choices even conditional on absolute achievement. Murphy and Weinhardt (2020) relate rank measured at the end of primary school to subsequent subject choices, while Delaney and Devereux (2021) study mathematics and English rank close to the transition to university. Elsner et al. (2021) show that, in a Dutch business school, a higher rank among randomly assigned section peers raises the probability of choosing a math-intensive elective, more for men than for women. Most closely related to our upper-secondary setting, Contini et al. (2025) examine realized track choice using absolute achievement, within-student differences across subjects, and class rank. Academic performance is strongly associated with track choice in their setting but changes the full-sample gender gap only modestly. Other work combines intra-individual and peer comparisons. Goulas et al. (2024), for example, rank students' relative STEM versus non-STEM performance within the classroom and relate this measure to realized STEM-track choice and subsequent STEM applications. This literature establishes that both intra-individual and peer-relative performance are relevant for educational specialization. It also motivates keeping the two constructs separate: when cross-subject performance is itself ranked within a peer group, their respective relationships with specialization cannot be distinguished. Moreover, because mathematics rank is constructed from the underlying mathematics score, it remains unclear how much of the gender gap associated with rank is distinct from the part associated with that score.

Accounting estimates in this literature vary widely and do not correspond to a common parameter. Studies differ in whether outcomes are stated intentions, applications, or realized enrollment; whether academic profiles are measured using standardized tests, grades, or latent skills; in the educational margin being studied; and in the definition of STEM. The gap used as the baseline also matters: in Ontario, students' within-cohort ranks in their STEM and overall course grades narrow the gender gap in STEM registration among STEM-ready university entrants from 5.8 to 1.7 percentage points, whereas 84% of the 13.2-point gap among all entrants is attributed to the gender gap in STEM-readiness rates (Card and Payne, 2021). These differences cannot be reduced to a simple distinction between intentions and realized choices: within the same US data, for example, Wan et al. (2023) obtain similar accounting shares for intended and realized college fields. Performance is also measured at different points in the educational trajectory. Early peer rank and an early mathematics-language differential have been related to

choices many years later (Murphy and Weinhardt, 2020; Denning et al., 2023; Hanushek et al., 2024), while other studies compare mathematics and language performance measured at different points within secondary schooling or substitute earlier measures of absolute achievement (Uunk, 2024; Contini et al., 2025). Cross-study differences in accounting shares therefore cannot by themselves be interpreted as differences in the underlying importance of relative academic performance. Existing evidence leaves particularly open how separately constructed within-student and peer-relative measures compare across successive specialization margins when the institutional setting, data source, and construction of the measures are held constant.

We address these questions by studying the same academic profiles across two successive specialization decisions using a common administrative data source. In Catalonia, students entering upper-secondary education (*Batxillerat*) choose among several tracks around age 16, one of which, the Science and Technology track (hereafter the Science track), is centered on mathematics and the natural sciences. Students who subsequently enter university make a second specialization decision when choosing a degree program. We construct within-student comparative advantage and peer-relative mathematics rank separately and relate them, alongside absolute mathematics achievement, to both realized choices. This design differs from Contini et al. (2025), who compare the three dimensions at a single realized track-choice margin, and from Goulas et al. (2024), who combine cross-subject and peer-relative performance in a single measure. Holding the institutional setting, data source, and construction of the measures fixed allows us to compare their accounting contribution across educational margins. It also allows us to distinguish whether a measure accounts for more of the gender gap because enrollment is more strongly associated with it or because girls and boys differ more in it, and to assess how much of the information contained in comparative advantage and peer rank overlaps with absolute mathematics achievement.

We link 10th-grade assessments, and for the 2000–2006 birth cohorts also 6th-grade assessments, to subsequent enrollment for 219,638 students. We measure comparative advantage in mathematics as the difference between standardized mathematics and reading scores and additionally construct students' peer-relative mathematics rank. Because students are assessed at two stages, we can also examine whether the timing of the performance measures matters for their accounting contribution. At each educational margin, we estimate the gender gap conditional on school, cohort, and student and family characteristics. The share accounted for by a performance measure is the percentage reduction in this gap when the measure is added. These exercises are descriptive rather than causal: they characterize how observed academic profiles are associated with the gender gap among students reaching each stage. In particular, the university sample is a selected subset of students who continue into higher education, and university enrollment additionally reflects admission.

Our first result is that comparative advantage accounts for more of the gender gap than absolute mathematics achievement, but primarily because girls and boys differ more in their relative mathematics and reading performance, not because, on the observed scores, enrollment

rises more steeply with comparative advantage. In the main samples, mathematics achievement alone accounts for 41.3% of the Science-track gap and 14.7% of the STEM-degree gap, compared with 48.5% and 18.5% for comparative advantage. When mathematics achievement and comparative advantage are considered jointly, the Gelbach (2016) decomposition assigns to comparative advantage beyond mathematics 5.3% of the original Science-track gap and 6.3% of the STEM-degree gap on the observed scores.

Second, the share accounted for by comparative advantage varies substantially across educational margins and with the definition of STEM. Among students observed in *Batxillerat*, and conditional on school, cohort, and family background, girls are 16.6 percentage points less likely than boys to enroll in the Science track, and 10th-grade comparative advantage accounts for 48.5% of this gap. Among university entrants, girls are 29.2 percentage points less likely to enroll in a STEM degree under our main definition, which includes the physical sciences, mathematics, computing, and engineering while excluding life sciences and medicine. Comparative advantage accounts for 18.5% of this gap. When life sciences and medicine are also classified as STEM, the gender gap falls to 18.6 percentage points and the share accounted for by comparative advantage rises to 38.0%. The apparent difference across stages is therefore substantially larger in proportional than in absolute terms: comparative advantage reduces the Science-track gap by 8.0 percentage points and the main university STEM gap by 5.4 points. Including life sciences and medicine matters because girls in the Science track frequently enter health-related programs, whereas boys rarely do (Figure 3).

Third, peer-relative mathematics rank is positively associated with STEM specialization conditional on mathematics achievement but adds little to the share of the gender gap already accounted for by the underlying score. Adding peer rank to mathematics achievement raises the share of the gap accounted for by less than half a percentage point at both margins. A small incremental accounting contribution of rank therefore does not imply that relative standing is unrelated to educational choices. Rather, in these data, much of the part of the gender gap associated with peer rank overlaps with the mathematics achievement from which rank is constructed.

The timing of the performance measures also matters for their accounting contribution. Comparative advantage measured in 6th grade accounts for 14.1% of the subsequent Science-track gap and 5.8% of the STEM-degree gap, substantially less than the corresponding 10th-grade measure. This comparison does not identify why the later measure accounts for more: the assessments are administered at different stages, may differ in measurement properties, and the 10th-grade measure is closer to the specialization decisions. The two educational margins also differ in how the conditional gender gap varies across the mathematics distribution. For Science-track enrollment, the gap changes relatively little across quartiles of 10th-grade mathematics. At university, by contrast, it rises on the probability scale from about 10 percentage points in the bottom quartile to more than 30 points in the top, a widening at the top that Breda et al. (2023) also document for stated intentions and, in auxiliary data, for realized enrollment. This widening

largely disappears when enrollment is modeled on a latent-index scale. We therefore interpret the latter pattern descriptively: a sizable conditional gender gap remains among students at the top of the mathematics distribution, without attributing the probability-scale widening to a particular behavioral mechanism.

Taken together, the findings qualify comparative advantage as a general explanation for women's underrepresentation in STEM. Relative academic strengths are systematically associated with specialization and account for a sizable share of the gender gap at some educational margins. Their accounting contribution, however, is not a stable property of the measure: it varies with the educational margin, the definition and size of the underlying STEM gap, and the stage at which performance is measured. Moreover, much of the contribution attributed to comparative advantage overlaps with absolute mathematics achievement. Our results therefore illustrate why estimates of the share of the STEM gender gap accounted for by comparative advantage can differ substantially across studies, and why those shares should not by themselves be interpreted as evidence for a distinct behavioral mechanism.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting and data. Section 3 presents the empirical framework and the interpretation of the accounting exercises. Section 4 reports the results, and Section 5 concludes.

2 Context and Data

In Catalonia, students entering academic upper-secondary education (*Batxillerat*, grades 11–12) choose a two-year track around age 16: Arts, Humanities and Social Sciences, or Science and Technology.[1] The Science track includes both STEM-oriented and health-related curricula. Track choice determines coursework and strongly shapes access to university programs.

University entry in Catalonia, as in the rest of Spain, further amplifies the importance of early specialization. Unlike in the US, students apply directly to specific degree programs and enroll in a field of study from the outset, rather than entering as undeclared and choosing a field later. Access to many STEM degree programs is tightly linked to completing the Science track. The track is not a formal requirement, but applicants are ranked by an admission score that adds optional subject exams, which each program weights by their relevance to it, so an engineering program rewards mathematics, a subject taught in the Science track. Consistent with this, very few students enter STEM degrees from other tracks: in our data, 97.2% of the STEM-degree entrants observed in *Batxillerat* had followed the Science track. Overall, the combination of track choice around age 16 and program-specific university entry implies that access to STEM is largely determined well before university enrollment.

[1] Since 2022–23, a General upper-secondary track has been introduced in some Catalan schools. This more flexible pathway combines elements from different specializations, but it is only available in a limited number of schools and was open only to the 2006 and 2007 birth cohorts. Almost none of the students observed in *Batxillerat* chose it, and they count as not enrolled in the Science track.

2.1 Data

We use administrative records from the Statistical Institute of Catalonia (Idescat) for a representative sample of students born between 2000 and 2007. The records link each student's test scores, school, family background, and enrollment in *Batxillerat* and university. The analysis sample includes 219,638 students with complete gender and family-background information and mathematics, Catalan, and Spanish scores in both 6th and 10th grade. Linked 6th-grade scores are available for the 2000–2006 cohorts.[2]

We study track choice among the 155,494 students observed in *Batxillerat*, 70.8% of the analysis sample. University records cover only the 2000–2003 birth cohorts; among their 75,516 *Batxillerat* students, 45,606 enter a Catalan university, providing the degree-choice sample.

We use three nested STEM definitions based on the Catalan Classification of Education Fields (CCED 2020; Appendix Table A.1). The main definition includes environmental sciences, chemistry, earth sciences, physics, mathematics, statistics, computing, engineering, and architecture. A second, broader definition corresponds to the fields the Science track prepares students for: it adds the life sciences, health programs, agriculture, forestry, and veterinary science to the main definition. The math-intensive definition retains the programs with the heaviest mathematics content.

Among *Batxillerat* students, girls are 16.7 percentage points less likely than boys to enroll in the Science track; among university entrants, they are 29.2 points less likely to enroll in a STEM degree (Figure 1). The university gap is smaller under the definition that also includes life sciences and medicine (Appendix Table A.3). Its larger size under the main definition does not reflect cohort coverage: the Science-track gap is 17.6 points in the cohorts observed at university. In the full analysis sample, the Science-track gap is only 5.5 points because more girls than boys continue to *Batxillerat* (77% versus 64%; Appendix Table A.3).

The Catalan education authorities administer standardized competency assessments to students in public and private schools at the end of 6th and 10th grade.[3] We use mathematics, Catalan and Spanish scores (Catalonia is a bilingual region), plus 10th-grade science scores.[4] The tests do not determine progression or graduation, although teachers may consider them when grading. The 10th-grade assessment precedes track choice by a few months. Reading is the average of Catalan and Spanish scores; using the higher score is a robustness check. We

[2]The 6th-grade test records delivered for the 2007 birth cohort are not linked to the students they belong to, so we set that cohort's 6th-grade scores, and the measures built from them, to missing: 27,319 of the 219,638 students. The cohort enters the descriptive statistics, the quartile analysis, and the peer-rank models; the Science-track models, which compare the two grades, use the students with scores in both, 137,134 of those observed in *Batxillerat*.

[3]The assessments are the *Proves de competències bàsiques a sisè de primària* (6th grade) and *a quart d'ESO* (10th grade), designed by the Consell Superior d'Avaluació del Sistema Educatiu, the evaluation agency of the Catalan education authorities.

[4]The extract lacks the 10th-grade science scores of the 2000 birth cohort, except for the 8% of its students who reached 10th grade late, a selected group; we leave that cohort out of every analysis that uses science.

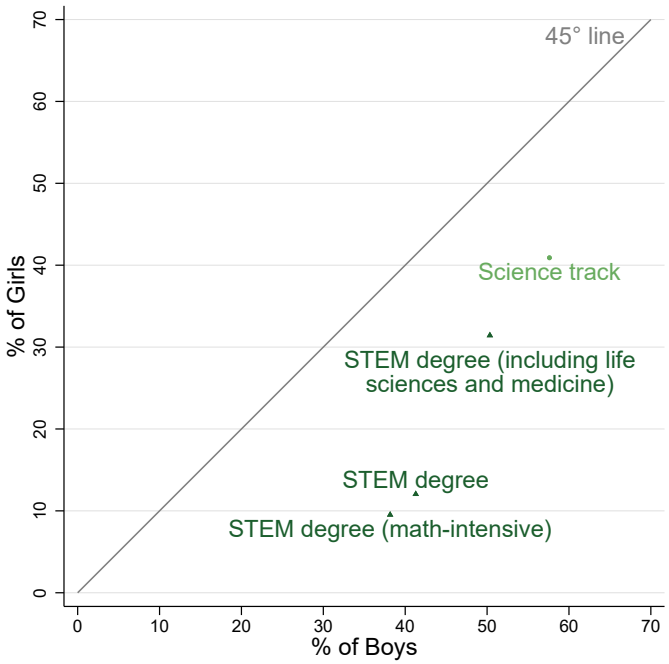


Figure 1: **STEM enrollment of girls and boys, by stage and definition.** Each marker is one outcome, with the share of girls enrolled in it on the vertical axis and the share of boys on the horizontal axis, each computed among all students of that gender who reach the stage, in percent. Science track: students observed in *Batxillerat* (N = 155,494). STEM degrees, under all three definitions: university entrants of the 2000–2003 birth cohorts observed in *Batxillerat* (N = 45,606). A marker on the 45-degree line would mean equal shares; every marker lies below it, so in each outcome a smaller share of girls than of boys enrolls. Source: Idescat administrative records; authors' calculations.

standardize scores within subject, grade, and cohort in the analysis sample because the means and dispersions of the original 0–100 scores vary across cohorts. Choice samples retain this standardization. From these scores we build our two main performance measures: within-student comparative advantage in mathematics (relative to reading) and peer rank in mathematics.

Our main measure of comparative advantage is $z_M - z_R$, the difference between standardized mathematics z_M and reading z_R scores, as in Sadowski and Zawistowska (2020); Breda and Napp (2019) standardize instead the difference between the raw scores, the construction we use as a robustness check. The standard deviation of $z_M - z_R$ is 0.84 in 6th grade and 0.88 in 10th grade. Both scores use cohort-wide rather than school-specific standardization, so the measure compares each student's relative performance across subjects. "Comparative advantage" refers to this measure unless stated otherwise.[5]

For peer rank, we measure 10th-grade mathematics rank among students observed in the same school and cohort as $(r-1)/(n-1)$, where r is the student's position and n the cell size, and standardize it for estimation (Elsner and Isphording, 2017; Murphy and Weinhardt, 2020).

[5] As a robustness check, we also standardize the raw difference between the two scores, z_{M-R}. It weights the standardized mathematics score by the ratio of the dispersions of the two raw scores, 1.74 in 10th grade (between 1.51 and 2.05 cohort by cohort), a property of the assessments rather than of students, so it is close to a second measure of mathematics achievement: in 10th grade it correlates 0.81 with the standardized mathematics score and 0.04 with the reading score. Appendix Table B.6 reports the results with this measure.

We also construct same-gender rank. To limit imprecision, both measures require school-cohort cells with more than 32 students and at least 10 of the student's gender. These restrictions yield 110,211 students, 50.2% of the analysis sample.[6] We do not construct 6th-grade ranks because the roughly 2,330 primary schools produce smaller cells than the 1,164 secondary schools.

The choice samples are positively selected on achievement but not on comparative advantage: students observed in *Batxillerat* score 0.2 to 0.3 standard deviations higher in mathematics and reading than the analysis sample, and university entrants about 0.2 standard deviations higher than students observed in *Batxillerat* in the same cohorts, while math comparative advantage differs by 0.05 or less between each choice sample and the sample it is drawn from.

2.2 Gender Differences in Performance Measures

While girls and boys in the analysis sample have very similar family background, their academic performance differs, in opposite directions across subjects (Appendix Table A.2). In 10th grade, boys score 0.25 standard deviations higher than girls in mathematics, and girls score 0.37 standard deviations higher in reading (Appendix Figure A.1). The two gaps therefore add up rather than cancel: girls' math comparative advantage is 0.61 below boys', more than either subject gap on its own. In science, boys score 0.10 standard deviations higher (Appendix Table A.2, Panel C), less than half the gap in mathematics.

The 10th grade math comparative advantage differs by gender even within mathematics deciles: at similar mathematics scores, girls score higher in reading which translates into a lower math comparative advantage (Figure 2). As a result, average comparative advantage turns positive in the fourth decile for boys but only in the eighth for girls, so from the fourth to the seventh decile the average boy's relative strength is already mathematics while the average girl's is still reading. Science-track enrollment is lower among girls in every decile and exceeds 50% only from the eighth decile, against the seventh for boys; among university entrants in the top decile, 26% of girls enroll in a STEM degree, against 59% of boys. These patterns are consistent with comparative advantage drawing boys toward STEM, and girls away from it, at the same mathematics score.

The gender difference in math comparative advantage grows from 0.39 in 6th grade to 0.61 in 10th grade (Appendix Table A.2), consistent with the age-related divergence in numeracy and literacy gaps documented by Borgonovi et al. (2021). Some of the increase may reflect assessment differences, including the concentration of 6th-grade mathematics scores near the ceiling. Comparative advantage is also less persistent across grades than either score: its correlation is 0.39, versus 0.68 for mathematics and 0.71 for reading (Appendix Table A.4). We therefore use 10th-grade measures as our main predictors and examine measurement-error sensitivity in Section 3.

[6] Tied scores receive their average position. The first floor alone leaves 110,348 students; the second excludes 137 more (0.1%), so that every column of a table that uses the ranks is estimated on the same students.

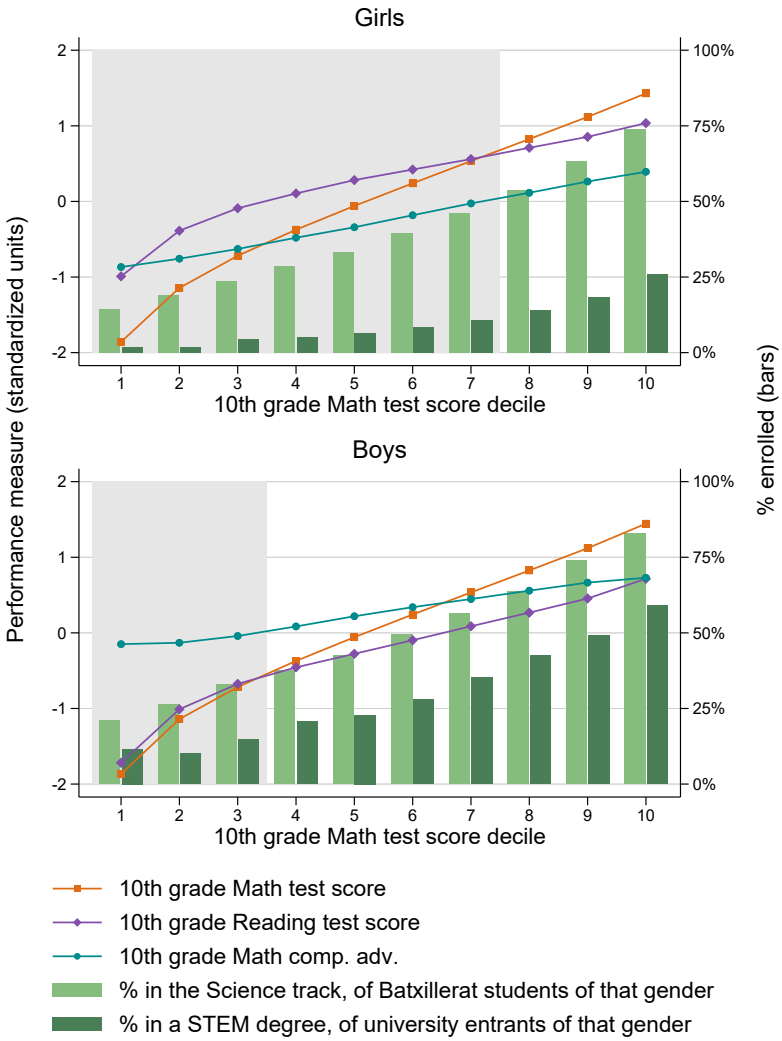


Figure 2: **10th-grade academic profiles and STEM enrollment by math decile, by gender.** Lines, left axis: average standardized math and reading scores and math comparative advantage (in units of the standardized scores), by decile of the 10th-grade math score, for girls (top panel) and boys (bottom panel) in the analysis sample (N = 219,638). The shaded region marks the deciles in which the average girl (top panel) or boy (bottom panel) has a comparative advantage in reading. Bars, right axis (0 to 100 percent), on the same deciles: the share of the students of that gender observed in *Batxillerat* (N = 155,494) who enroll in the Science track (light green), and the share of the university entrants of that gender, from the 2000–2003 birth cohorts (N = 45,606), who enroll in a STEM degree under the main definition (dark green). Source: Idescat administrative records; authors' calculations.

Gender differences in peer rank are smaller than in comparative advantage. The normalized gender difference is 0.26 for both rank and mathematics achievement, versus 0.76 for math comparative advantage (Appendix Table A.5). Girls' average rank is 0.46, compared with boys' 0.54; their share falls from 60.4% in the lowest rank decile to 40.3% in the highest (Appendix Figure A.2). Rank and comparative advantage correlate at 0.43 and classify 33.0% of students differently: their rank is above the cohort median but their comparative advantage favors reading, or vice versa. The first combination is almost twice as common among girls as boys (21.3% versus 11.8%; Appendix Table A.6).

3 Empirical Framework

We relate STEM specialization to students' academic profiles with linear probability models, estimated separately for each choice:

$$Y_{isc} = \alpha \, Female_i + \beta A_i + X_i'\delta + \lambda_s + \gamma_c + \varepsilon_{isc}, \tag{1}$$

Here, Y_{isc} indicates Science-track enrollment among *Batxillerat* students or STEM-degree enrollment. $Female_i$ identifies girls, and A_i is one performance measure at a time: the mathematics score, reading score, or math comparative advantage in 6th or 10th grade, the 10th-grade science score, or peer rank in mathematics (among all or same-gender schoolmates). Controls X_i include student and parental foreign birth, household structure, and parental university education (Appendix Table A.2).[7] School and cohort fixed effects are denoted by λ_s and γ_c; standard errors are clustered by 10th-grade school. The coefficient α is the conditional gender gap, and β the enrollment slope on performance. Both are descriptive because unobserved preferences, beliefs, and other determinants may correlate with gender and achievement.

We use additive school and cohort fixed effects because school-by-cohort effects would identify the gap only within cells containing both genders, changing the comparisons underlying the estimates. Among university entrants, half of these cells contain fewer than 10 students and 12.8% contain only one gender. Clustering by school allows shocks to persist across cohorts; both samples include more than 1,000 schools.

Because our question is what students choose to study rather than whether they continue studying, we estimate each model among the students who reach the corresponding stage. Outcomes defined for the full analysis sample combine continuation and field choice, which work in opposite directions for girls: they are more likely to enter *Batxillerat* but less likely to choose Science once there. Appendix C reports these overall-participation outcomes.

Since math comparative advantage is the difference between the mathematics and reading scores, the three measures cannot enter one model; we add them one at a time. Let α_0 and

[7] Year of birth is omitted because it coincides with the birth cohort for 96.1% of students, and whether a school is publicly or privately owned is absorbed by the school fixed effects.

α_A denote α in the models without and with A_i, estimated on the same students, and β_A the corresponding β. The quantity of interest is the share of the gender gap that a measure accounts for, $100(\alpha_0 - \alpha_A)/\alpha_0$. By the omitted-variable formula, $\alpha_0 - \alpha_A = \beta_A \theta_A$, where θ_A is the coefficient on $Female_i$ in the auxiliary regression $A_i = \theta_A Female_i + X_i'\delta^A + \lambda_s^A + \gamma_c^A + u_{isc}$ on the same students. The share therefore depends on the conditional gender difference in performance, θ_A, its enrollment slope, β_A, and the baseline gap, α_0. It is zero if either of the first two components is zero, and a given percentage-point reduction accounts for more of a smaller gap.

Appendix Table B.5 reports the three components. This accounting exercise does not identify mediation or the gap reduction that would follow an intervention on academic profiles. For 10th-grade comparative advantage, we obtain 95% confidence intervals by resampling 10th-grade schools.

To separate comparative advantage from mathematics achievement, we also enter the 10th-grade mathematics score and comparative advantage together. Once z_M is held fixed, $z_M - z_R$ varies only with the reading score, so this joint model is a reparameterization of one with the two scores, and it relaxes the constraint, implicit in entering $z_M - z_R$ alone, that their slopes are equal and opposite. Following Gelbach (2016), we split the resulting change in the gap as $\alpha_0 - \alpha_{M,CA} = \beta_M \theta_M + \beta_{CA} \theta_{CA}$, where $\alpha_{M,CA}$, β_M, and β_{CA} are the coefficients on $Female_i$, z_M, and $z_M - z_R$ in the joint model and the θs come from the auxiliary regressions above. The split does not depend on the order in which the measures are added but does depend on the pair, so we also report it for the mathematics and reading scores.

We examine heterogeneity by interacting $Female_i$ with each performance measure and, separately, with quartiles of 10th-grade mathematics defined in the analysis sample.

Peer rank enters the model in the same way as the other measures, in the school-cohort cells in which it is defined (Section 2.1): first alone, then together with the mathematics score, as in the rank literature, and with a cubic in the score. Within a school-cohort cell, the rank is a monotone transformation of the score, so these models describe how the two measures are related to choices; we do not use variation in the composition of school cohorts to identify an effect of rank.

Three features limit interpretation. First, school fixed effects absorb time-invariant differences across schools, including course offerings, but not within-school differences in preferences, beliefs, or encouragement. Second, reaching each stage selects students differently by gender and achievement, so cross-stage gaps need not measure the same parameter. Among *Batxillerat* students, 62.9% of girls and 57.3% of boys enter university. A Lee-style trimming exercise (Lee, 2009) bounds the raw STEM-degree gap between -37.8 and -28.1 percentage points, while inverse-probability reweighting on observed characteristics leaves the comparative-advantage share nearly unchanged (Appendix Tables B.7 and B.8). Third, university enrollment reflects both applications and admission.

Measurement error introduces further uncertainty. Comparative advantage combines errors from two correlated scores and may therefore be less reliable than either score, particularly when mathematics scores are concentrated near the ceiling. Rank also inherits error from the mathematics score. Because assessment reliabilities are unpublished, Appendix Table B.9 examines classical measurement error under assumed score reliabilities and error correlations. Under these assumptions, correcting the separately entered mathematics and comparative-advantage measures increases their accounting shares, while the allocation in the joint model changes substantially. Section 4.2 illustrates this sensitivity at a reliability of 0.90.

4 Results

All models follow Section 3 unless stated otherwise. In Sections 4.1 to 4.3, except for the quartile analysis, Science-track estimates use the common sample with scores in both grades: 137,128 *Batxillerat* students after excluding singleton fixed-effect groups. This excludes the 2007 cohort; the university sample is unaffected.[8]

4.1 The Gender Gap Accounted for by Academic Profiles

The conditional Science-track gap is 16.6 percentage points. Adding 10th-grade mathematics achievement reduces it by 41.3%, to 9.7 points; adding comparative advantage reduces it by 48.5%, to 8.5 points (95% confidence interval: 46.6–50.8%). Appendix Table B.1 reports the full estimates, and Appendix Table B.5 collects the gender gap with each measure and the share of the gap each measure accounts for. Reading alone widens the gap to 19.4 points because girls score higher and reading positively predicts enrollment when mathematics is omitted. Its conditional association reverses when mathematics is included. The 6th-grade measures account for less: 21.8% for mathematics and 14.1% for comparative advantage.

Among university entrants, the conditional STEM gap is 29.2 percentage points. Mathematics achievement reduces it by 14.7%, to 24.9 points, and comparative advantage by 18.5%, to 23.8 points (95% confidence interval: 17.4–19.7%); reading alone slightly widens it. As for the Science track, Appendix Table B.2 reports the full estimates, and Appendix Table B.5 the gap and the share accounted for by each measure. Under the definition that also includes life sciences and medicine, the gap is 18.6 points and the respective shares are 35.7% and 38.0% (Appendix Table B.5); the math-intensive results resemble the main estimates (Appendix Table B.4). Appendix Table B.6 reports the robustness checks for every enrollment outcome. Sixth-grade mathematics and comparative advantage account for only 7.7% and 5.8%. At both stages, all performance slopes are larger in 10th grade (Appendix Table B.5).

[8] On all students observed in *Batxillerat*, the 10th-grade estimates are close: the conditional gap is 16.2 percentage points, and comparative advantage accounts for 49.6% of it.

The accounting identity explains both comparisons (Appendix Table B.5). Girls' conditional comparative advantage is 0.57 standardized-score units below boys' in *Batxillerat* and 0.50 below among university entrants; the corresponding slopes are 0.141 and 0.109. The reductions are therefore 8.0 and 5.4 percentage points, with the larger university gap driving much of the difference in shares. Within *Batxillerat*, comparative advantage accounts for more than mathematics despite its flatter slope (0.141 versus 0.210) because its gender difference is larger (0.57 versus 0.33).

The larger 10th-grade Science-track share reflects both a steeper comparative-advantage slope (0.141 versus 0.069) and a larger conditional gender difference (0.57 versus 0.34). The latter mainly reflects the increase in girls' reading advantage, from 0.08 to 0.25, rather than their mathematics disadvantage, from 0.26 to 0.33 (Appendix Table B.5).

Figure 3 traces specialization in the cohorts observed at university. Among *Batxillerat* students, 40% of girls and 58% of boys choose Science; 8% of girls subsequently enter STEM degrees and another 9% health programs, compared with 24% and 3% of boys. Among university entrants, track composition accounts for 27.3% of the raw 29.2-point STEM gap under the symmetric decomposition, or 16.0–38.6% using girls' or boys' enrollment rates as reference. Of the remaining 21.2 points, 20.3 arise within the Science track (Appendix Table B.10). Adding track indicators reduces the conditional gap from 29.2 to 22.1 points; within Science, comparative advantage accounts for 10.8%, versus 18.5% among all entrants. Health enrollment is consistent with the larger accounting shares under the life-sciences and medicine definition. These decompositions locate the gap rather than identify its causes.

Results are similar across fixed-effect specifications, clustering choices, and definitions of reading (Appendix Table B.6). School-by-cohort fixed effects yield comparative-advantage shares of 50.4% and 19.3%, versus 48.5% and 18.5% with additive effects. Construction matters more: the shares are 61.4% and 21.3% using the standardized raw-score difference z_{M-R}, and 43.9% and 17.4% using a percentile-rank difference.[9]

4.2 Comparative Advantage beyond Mathematics Achievement

Entering 10th-grade mathematics achievement and comparative advantage jointly reduces the Science-track gap by 45.0%, to 9.1 percentage points, and the STEM-degree gap by 18.6%, to 23.8 points. The Science-track share is smaller than with comparative advantage alone (48.5%) because the accounting share measures the change in the gender coefficient, not model fit. Allowing separate mathematics and reading slopes improves fit without necessarily reducing that coefficient further. The Gelbach (2016) decomposition allocates 88.2% of the Science-track reduction to mathematics and 11.8% to comparative advantage (95% confidence interval

[9]Defined with the standardized science score in place of the mathematics score, comparative advantage accounts for less of each gap: on the students with a 10th-grade science score, 35.9% of the Science-track gap and 14.4% of the STEM-degree gap, against 51.4% and 19.2% for math comparative advantage.

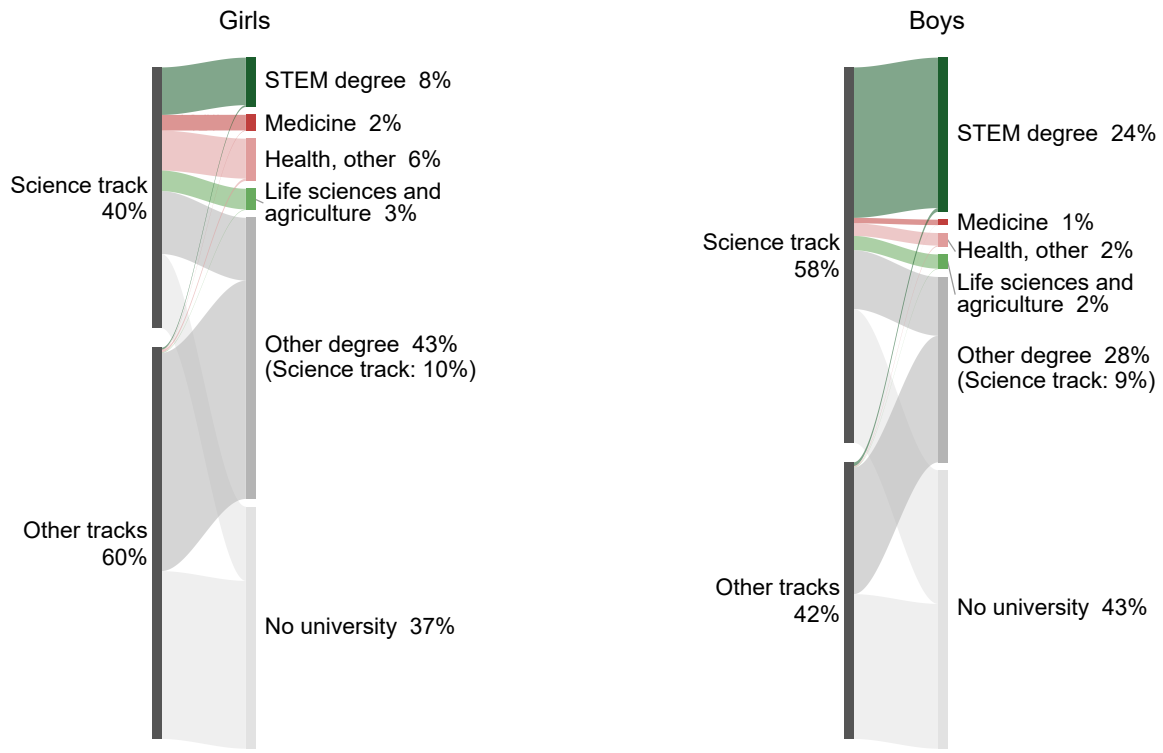


Figure 3: **From *Batxillerat* track to university, by gender.** Flows link *Batxillerat* tracks to university destinations, separately by gender. Percentages use each gender's *Batxillerat* cohort as denominator; destinations are mutually exclusive and exhaustive. STEM uses the main definition. Medicine is shown separately from other health programs; life sciences are grouped with agriculture, forestry, and veterinary science. Sample: 2000–2003 birth cohorts observed in *Batxillerat* with a 10th-grade school identifier (41,888 girls and 33,628 boys). Source: Idescat administrative records; authors' calculations.

for the latter: 8.9–14.8%). At university, the corresponding allocations are 66.0% and 34.0% (28.3–38.8%; Appendix Table B.5, Panel E). The comparative-advantage components represent 5.3% and 6.3% of the original gaps. These are allocations within the joint model, not the sequential increases obtained by adding comparative advantage to mathematics.

The allocation depends on parameterization and measurement error. Expressing the same model in mathematics and reading scores assigns 94.9% and 87.8% of the reductions to mathematics. Under the classical-error assumptions in Appendix Table B.9, setting both score reliabilities to 0.90 raises the comparative-advantage allocations from 5.3% and 6.3% of the gaps to 23.2% and 17.0%. The observed-score results therefore show substantial overlap, while the precise allocation remains sensitive to measurement assumptions.

4.3 Slopes by Gender and the Gap across the Mathematics Distribution

In the Science track, girls and boys have statistically indistinguishable slopes on our main comparative-advantage measure (interaction 0.004, standard error 0.003; Appendix Table B.11). At university, girls' slope is about half of boys' (0.082 versus 0.148). This contrast depends on scale: STEM enrollment rates are 12.0% for girls and 41.3% for boys (Appendix Table A.3), and the latent-index interaction is positive in both a probit controlling for school-by-cohort regressor means (0.048, standard error 0.021) and a conditional logit within school-cohort cells (0.198, standard error 0.038).

The conditional Science-track gap rises from 7.6 percentage points in the bottom mathematics quartile to 10.8 in the top; the STEM-degree gap rises from 10.2 to 32.6 points (Appendix Table B.11). University entrants are concentrated in the upper quartiles, making bottom-quartile estimates less precise. The widening is much smaller on the probit latent-index scale: the university gap changes from 0.85 to 0.90 and the Science-track gap from 0.26 to 0.32. At both stages, index gaps are statistically indistinguishable across the bottom three quartiles; the top-quartile gap is larger than the bottom one in the Science track, and at university it is larger only than the third. Index units are comparable across quartiles within each stage, but not across stages.

Under the trimming assumptions in Appendix Table B.7, differential university entry does not appear to account for the widening under the main STEM definition: even combining the most adverse within-quartile bounds, a descriptive exercise rather than a bound on the difference, leaves the top-quartile raw gap 17.9 percentage points above the bottom-quartile gap, versus 22.1 points without trimming. Under the definition that also includes life sciences and medicine, however, the bounds allow a smaller top-quartile gap.

4.4 Peer Rank in Mathematics

Peer rank accounts for similar shares of the gaps as mathematics achievement. Comparing all performance measures requires a science score, restricting the rank sample to the 2001–2007 cohorts; scores and ranks are restandardized in this sample (Appendix Tables B.12 and B.13). The conditional gaps are 15.9 percentage points for Science and 29.6 for university STEM. Rank accounts for 44.5% and 14.2%, close to mathematics achievement's 46.2% and 14.4%. Comparative advantage accounts for 54.4% and 19.0%, and science achievement for 23.3% and 8.7%. Rank is constructed from the mathematics score and inherits its measurement error, limiting interpretation of their separate contributions.

Figure 4 separates each measure's enrollment slope from its conditional gender difference. Comparative advantage accounts for more than mathematics despite a flatter slope because its standardized gender difference is larger (0.60 versus 0.34). Same-gender rank accounts for only 11.6% and 5.5% of the gaps despite a steep slope: ranking each gender separately largely removes the gender difference by construction. Residual differences among students reaching each stage reflect selection.

On the rank sample without the science-score restriction, rank remains positively associated with enrollment conditional on mathematics achievement: a standard-deviation increase corresponds to 9.8 percentage points for Science and 7.0 for university STEM, while mathematics slopes fall by about half (Appendix Tables B.14 and B.15). Yet adding rank barely changes the accounting shares, from 45.4% to 45.8% and from 14.8% to 15.1%. A cubic mathematics specification raises them only to 46.7% and 15.6%.

5 Conclusion

This paper asks how two reference points for students' mathematics performance, their own performance in reading and their standing among schoolmates, relate to the gender gap in STEM specialization at two stages, the choice of the Science track in upper-secondary education and the choice of a STEM degree at university, using administrative records for eight birth cohorts in Catalonia and comparing girls and boys conditional on school, cohort, and family background.

Four results stand out. First, 10th-grade math comparative advantage accounts for about half of the conditional gender gap in Science-track enrollment but for less than a fifth of the gap in STEM-degree enrollment, or 38.0% when STEM also includes life sciences and medicine, a contrast that comes mainly from the size of the university gap: in percentage points, the two reductions, 8.0 and 5.4, are much closer than the shares. Second, it accounts for more than the mathematics score because girls and boys differ more in it, and much of what it accounts for is shared with the mathematics score; how the reduction divides between the two cannot

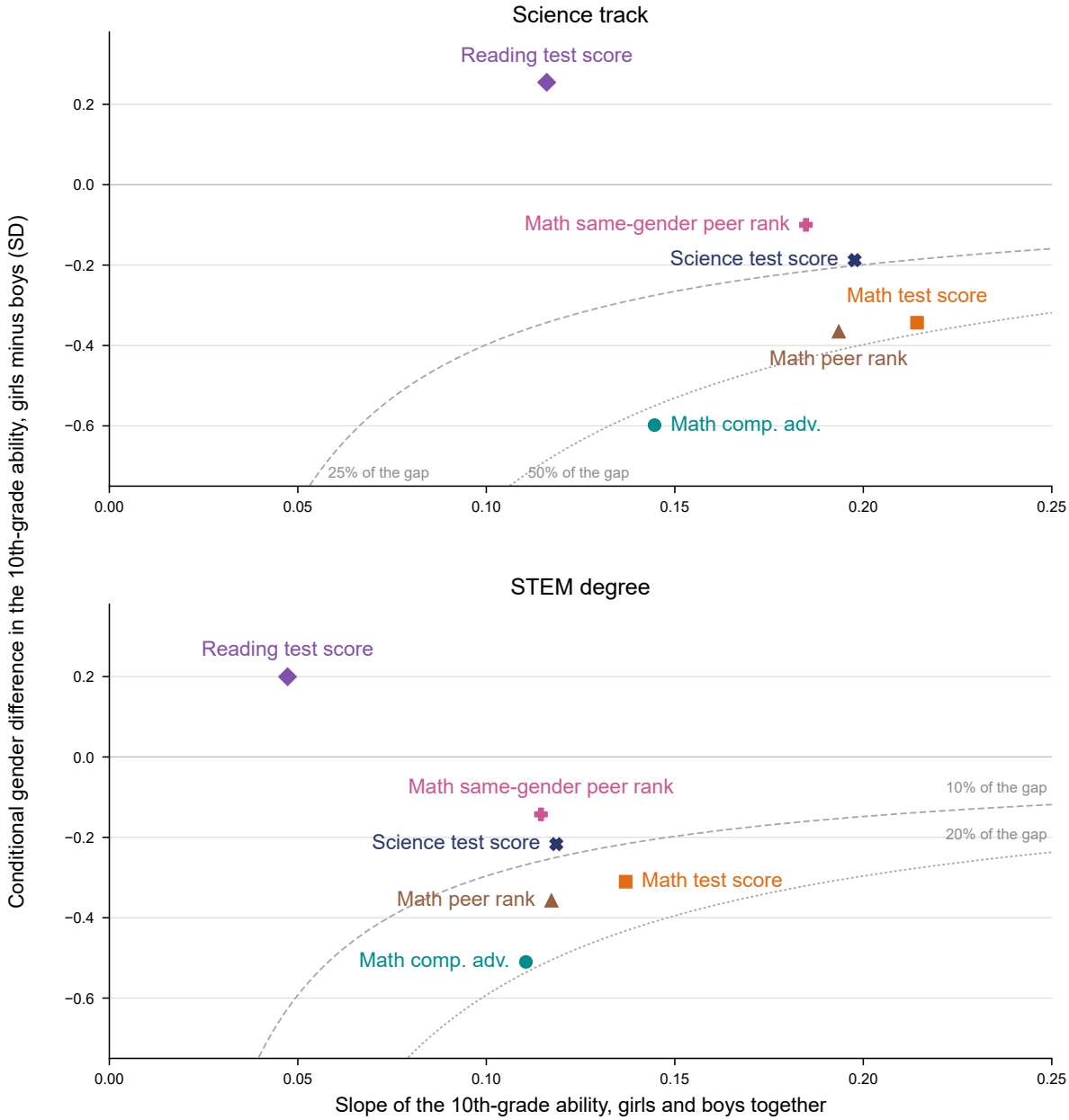


Figure 4: **Why one performance measure accounts for more of the gender gap than another.** Each marker plots a performance measure's enrollment slope, β_A (horizontal axis), against its conditional gender difference, θ_A (vertical axis). Curves identify equal accounting shares, $100\beta_A\theta_A/\alpha_0$, using baseline gaps of 0.159 for Science and 0.296 for university STEM. All six measures use a common sample within each panel: *Batxillerat* students from the 2001–2007 cohorts with science scores and eligible school-cohort cells (N = 68,095); university entrants among them belong to the 2001–2003 cohorts (N = 15,691). Cells require more than 32 students and at least 10 of the student's gender. Scores and ranks are standardized within cohort in this sample. Coefficients appear in Appendix Tables B.12 and B.13. Source: Idescat administrative records; authors' calculations.

be pinned down without the reliability of the assessments, and peer rank, constructed from the mathematics score, inherits its measurement error. Third, the two margins differ in how choices relate to academic profiles: at the choice of track, girls' and boys' enrollment rises at similar rates with comparative advantage and the gap changes relatively little across the mathematics distribution, whereas at university girls' enrollment rises about half as steeply as boys' with comparative advantage and the gap more than triples between the bottom and top quartiles of mathematics. Both statements are about percentage points: on the scale of a latent index, the widening largely disappears and girls' response to comparative advantage is, if anything, the stronger. Fourth, the gender difference in comparative advantage widens between 6th and 10th grade, and the 10th-grade measures account for much more of both gaps, 48.5% against 14.1% of the Science-track gap.

These results have limits. They are associations on selected samples: each choice is observed only among students who reach that stage, more girls than boys reach *Batxillerat*, and university enrollment also reflects applications and admission. University outcomes rest on four birth cohorts and on enrollment in Catalan universities. Because a difference of two scores carries the measurement error of both, the shares that the mathematics score and comparative advantage account for are lower bounds under classical error, and the part attributed to comparative advantage beyond mathematics could be several times larger at plausible reliabilities. And because the 6th- and 10th-grade assessments differ and the later one is closer to the decision, the larger 10th-grade shares do not show that comparative advantage formed in secondary school matters more for choices.

What the results do show is where explanations beyond academic profiles are most needed. That comparative advantage accounts for a large share of the gap mainly reflects how much girls and boys differ in it, and much of that share is also accounted for by the mathematics score. At the choice of track, differences in academic profiles account for a large part of the gender gap. At university, most of the gap remains among girls and boys with the same observed academic profile, and it is largest among the students best prepared in mathematics.

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Appendix A. Additional Descriptive Evidence

This appendix collects the descriptive evidence behind Section 2. Appendix Table A.1 lists the fields of study in each STEM definition, Appendix Table A.2 compares the backgrounds and scores of girls and boys, and Appendix Table A.3 reports continuation and STEM enrollment by gender. Appendix Tables A.4 to A.6 report the persistence of the measures between 6th and 10th grade, and peer rank by gender and its agreement with comparative advantage. Appendix Figure A.1 shows the score distributions, and Appendix Figure A.2 the share of girls by decile of peer rank in math.

Field of study	STEM degree (math-intensive)	STEM degree	STEM degree (including life sciences and medicine)
In all three definitions			
Chemistry	✓	✓	✓
Earth sciences	✓	✓	✓
Physics	✓	✓	✓
Mathematics	✓	✓	✓
Statistics	✓	✓	✓
Natural sciences, mathematics and statistics, not further defined	✓	✓	✓
Software and applications development and analysis	✓	✓	✓
Information and communication technologies, not elsewhere classified	✓	✓	✓
Information and communication technologies, not further defined	✓	✓	✓
Chemical engineering and processes	✓	✓	✓
Environmental protection technology	✓	✓	✓
Electricity and energy	✓	✓	✓
Electronics and automation	✓	✓	✓
Mechanics and metal trades	✓	✓	✓
Motor vehicles, ships and aircraft	✓	✓	✓
Engineering, not elsewhere classified	✓	✓	✓
Building and civil engineering	✓	✓	✓
Engineering, manufacturing and construction, not further defined	✓	✓	✓
Not in the math-intensive definition			
Environmental sciences	✗	✓	✓
Computer use	✗	✓	✓
Database and network design and administration	✗	✓	✓
Architecture and town planning	✗	✓	✓
Only in the definition including life sciences and medicine			
Biology	✗	✗	✓
Biochemistry	✗	✗	✓
Natural environments and wildlife	✗	✗	✓
Crop and livestock production	✗	✗	✓
Forestry	✗	✗	✓
Veterinary	✗	✗	✓
Dental studies	✗	✗	✓
Medicine	✗	✗	✓
Nursing and midwifery	✗	✗	✓
Therapy and rehabilitation	✗	✗	✓
Pharmacy	✗	✗	✓
Health, not elsewhere classified	✗	✗	✓

Table A.1: **Fields of study in each definition of a STEM degree.** Fields of the Catalan Classification of Education Fields (CCED 2020) included (✓) or not included (✗) in each definition of a STEM degree: math-intensive, main, and including life sciences and medicine. The definitions are nested. Fields not listed belong to none of them. Source: Idescat administrative records; authors' calculations.

	Girls	Boys	Difference (boys − girls)
Panel A. Student, family, and school characteristics			
Foreign-born student	0.07	0.06	-0.01^{***}
			(0.001)
Two-parent household	0.90	0.91	0.01^{***}
			(0.002)
Same-sex parents	0.012	0.012	−0.001
			(0.0005)
All parents with a university degree	0.19	0.20	0.01^{**}
			(0.003)
One parent with a university degree	0.21	0.22	0.01^{***}
			(0.002)
One foreign-born parent	0.095	0.091	-0.003^{**}
			(0.0013)
Both parents foreign-born	0.11	0.11	-0.01^{***}
			(0.001)
Public school, 10th grade	0.62	0.61	−0.01
			(0.005)
Panel B. Scores in 6th grade			
6th grade Math test score	−0.10	0.11	0.21^{***}
			(0.005)
6th grade Math test score (0–100)	78.85	82.05	3.21^{***}
			(0.076)
6th grade Reading test score	0.09	−0.09	-0.18^{***}
			(0.005)
6th grade Reading test score (0–100)	77.46	75.18	-2.28^{***}
			(0.067)
6th grade Math comp. adv.	−0.19	0.20	0.39^{***}
			(0.004)
Panel C. Scores in 10th grade			
10th grade Math test score	−0.12	0.13	0.25^{***}
			(0.005)
10th grade Math test score (0–100)	66.18	71.00	4.82^{***}
			(0.108)
10th grade Reading test score	0.18	−0.19	-0.37^{***}
			(0.006)
10th grade Reading test score (0–100)	79.37	75.26	-4.11^{***}
			(0.062)
10th grade Math comp. adv.	−0.30	0.31	0.61^{***}
			(0.004)
Observations	111,376	108,262	
10th grade Science test score	−0.05	0.05	0.10^{***}
			(0.005)
10th grade Science test score (0–100)	66.05	67.71	1.67^{***}
			(0.089)
Observations	97,232	94,836	

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.2: **Student and family background, school characteristics, and academic performance, by gender.** Means for girls and boys in the analysis sample (N = 219,638), and their difference. Rows whose difference would round to 0.00 are shown with three decimals. Panel B covers the 2000–2006 birth cohorts, those that contribute 6th-grade scores (N = 192,319). In Panel C, the science rows restrict the sample to the 192,068 students of the 2001–2007 birth cohorts with a 10th-grade science score; the 2000 birth cohort is excluded because the extract lacks its 10th-grade science scores. Rows other than test scores are proportions. Standardized scores have mean zero and standard deviation one within subject, grade, and cohort in the analysis sample; raw scores range from 0 to 100. Reading is the average of the Catalan and Spanish scores. Math comparative advantage is the difference between a student's standardized math and reading scores; its standard deviation is 0.84 in 6th grade and 0.88 in 10th. Parents refers to parents or legal guardians; a university degree excludes higher-level vocational training; public schools exclude publicly funded private (*concertada*) schools. Source: Idescat administrative records; authors' calculations.

	Girls	Boys	Difference (boys − girls)
Panel A. Continuation to upper-secondary education and track choice			
Continues to Batxillerat	77.1	64.3	-12.8^{***}
			(0.31)
Observations	111,376	108,262	
Among students observed in Batxillerat			
Science track	40.9	57.6	16.7^{***}
			(0.31)
Observations	85,902	69,592	
Panel B. Continuation to university and degree choice, 2000–2003 birth cohorts			
Enters university	50.6	38.8	-11.8^{***}
			(0.39)
Observations	53,055	50,897	
Among university entrants observed in Batxillerat			
STEM degree	12.0	41.3	29.2^{***}
			(0.44)
STEM degree (including life sciences and medicine)	31.4	50.3	18.9^{***}
			(0.48)
STEM degree (math-intensive)	9.5	38.2	28.6^{***}
			(0.42)
Observations	26,342	19,264	

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.3: **Continuation and STEM enrollment, by gender.** Share of girls and of boys, in percent, and their difference in percentage points, computed from unrounded shares. Panel A: continuation to *Batxillerat* in the analysis sample, and enrollment in the Science track among students observed in *Batxillerat*. Panel B covers the 2000–2003 birth cohorts, whose university enrollment is observed (2018–19 to 2021–22): entry into university, and enrollment in a STEM degree among university entrants observed in *Batxillerat*. The fields of study in each definition are listed in Appendix Table A.1. Source: Idescat administrative records; authors' calculations.

	Girls	Boys	All	p-value: girls = boys
Math test score	0.678	0.665	0.676	<0.001
Reading test score	0.727	0.699	0.713	<0.001
Math comp. adv. ($z_M - z_R$)	0.331	0.338	0.385	0.075
Math comp. adv. (z_{M-R})	0.347	0.313	0.373	<0.001
Observations	97,623	94,696	192,319	

Table A.4: **Correlation between 6th- and 10th-grade performance measures, within student.** Pearson correlation between each student's 6th- and 10th-grade values, for girls, boys, and all students, and the p-value of a test that the correlations of girls and boys are equal (Fisher transformation). The last two rows are math comparative advantage under $z_M - z_R$ and its alternative construction, z_{M-R}. The pooled correlation of comparative advantage exceeds the within-gender ones partly because the means of girls and boys differ in both grades. Analysis-sample students with scores in both grades, N = 192,319 (97,623 girls and 94,696 boys); the 2007 birth cohort is excluded because its 6th-grade test records are not linked to the students (Section 2.1). Source: Idescat administrative records; authors' calculations.

	Girls	Boys	Difference (boys − girls)	Normalized difference
10th grade Math test score	−0.11	0.14	0.25***	0.26
			(0.008)	
10th grade Math comp. adv.	−0.30	0.32	0.63***	0.76
			(0.005)	
10th grade Math peer rank	0.46	0.54	0.07***	0.26
			(0.002)	
Observations	55,830	54,381		

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.5: **Peer rank in math, math score, and comparative advantage, by gender.** Means for girls and boys and their difference. Peer rank is the percentile rank in 10th-grade math among the students in the data from the same school and cohort, from 0 (lowest) to 1 (highest). Appendix Table A.6 reports how far the two signals agree. Sample: students in school-cohort cells with more than 32 students in the data and at least 10 students of their own gender (N = 110,211). Source: Idescat administrative records; authors' calculations.

	Girls	Boys	All
Correlation between 10th grade Math peer rank and 10th grade Math comp. adv.	0.483	0.346	0.429
Share whose peer rank and comp. adv. disagree	0.329	0.332	0.330
of which: above the cohort median in Math, comp. adv. in Reading	0.213	0.118	0.166
of which: below the cohort median in Math, comp. adv. in Math	0.116	0.215	0.165
Observations	55,830	54,381	110,211

Table A.6: **Agreement between the two reference points: math comparative advantage and peer rank.** Correlation between the peer rank in 10th-grade math and math comparative advantage, and the share of students whose two signals point in opposite directions: they rank in the upper half of their school cohort while having a comparative advantage in reading, or in the lower half while having one in math. The last three rows are shares of students, between 0 and 1. Sample: students in school-cohort cells with more than 32 students in the data and at least 10 students of their own gender (N = 110,211; 55,830 girls and 54,381 boys). Source: Idescat administrative records; authors' calculations.

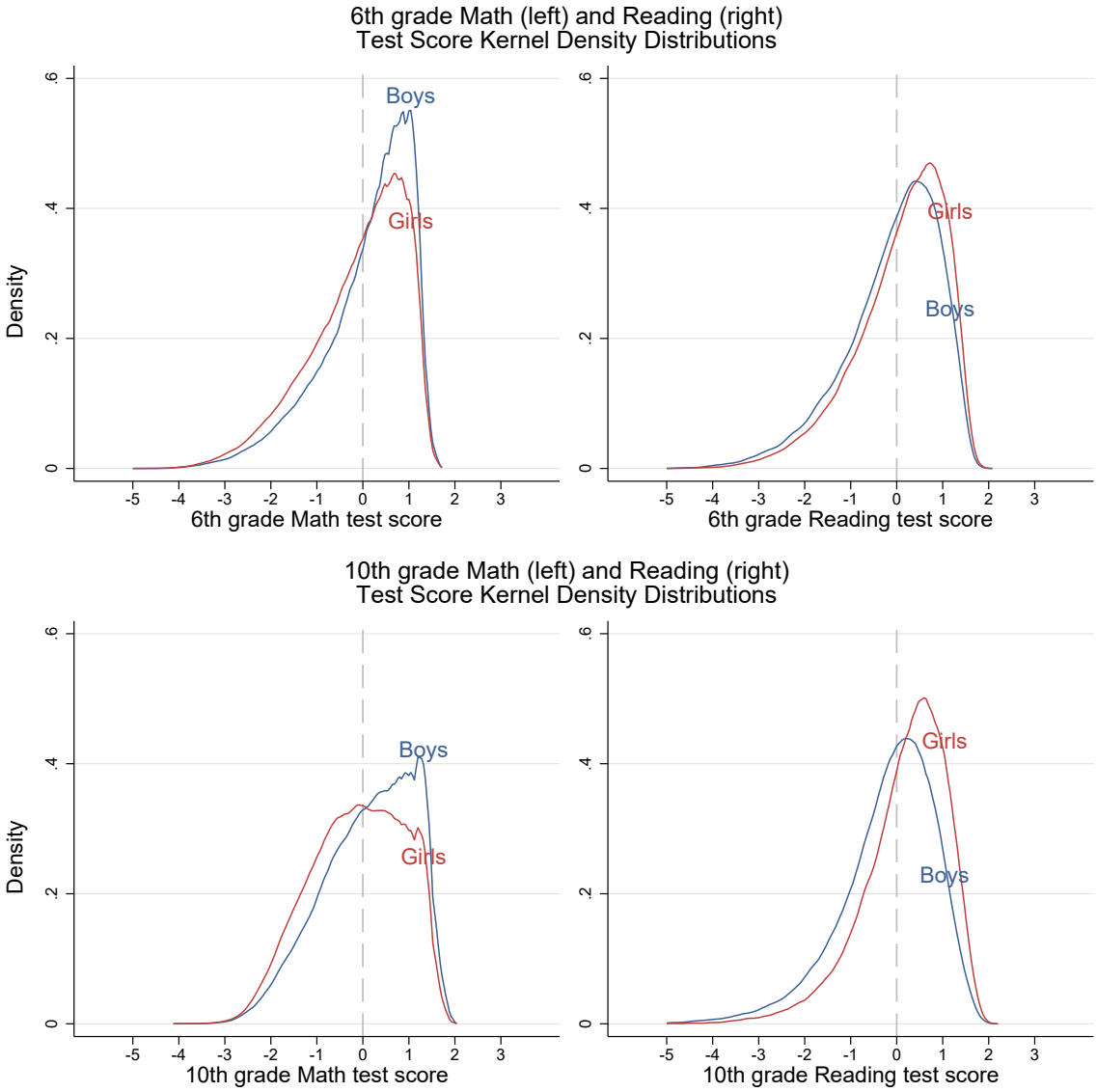


Figure A.1: **Distribution of math and reading scores in 6th and 10th grade, by gender.** Kernel densities of standardized math (left) and reading (right) scores in 6th grade (top) and 10th grade (bottom), for girls and boys, estimated with an Epanechnikov kernel. Analysis sample (N = 219,638 in 10th grade and 192,319 in 6th grade, which excludes the 2007 birth cohort). Source: Idescat administrative records; authors' calculations.

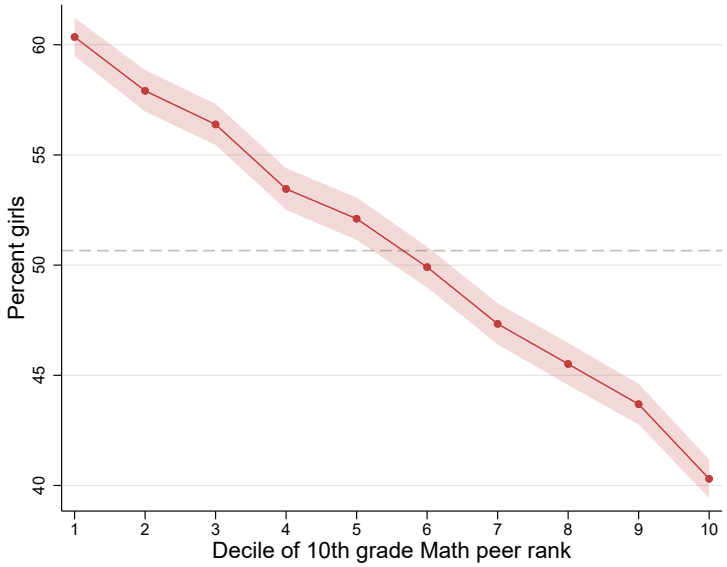


Figure A.2: **Share of girls by decile of peer rank in math.** Percentage of girls among students in each decile of peer rank in 10th-grade math, the percentile rank among the students in the data from the same school and cohort, with 95% confidence intervals. Deciles are fixed intervals of the rank, from [0, 0.1) to [0.9, 1]. The dashed line marks the share of girls in the sample (50.7%). Sample: students in school-cohort cells with more than 32 students in the data and at least 10 students of their own gender (N = 110,211). Source: Idescat administrative records; authors' calculations.

Appendix B. Additional Results

This appendix reports the full estimates behind Section 4. Appendix Tables B.1 to B.4 report the models for the Science track and the three STEM definitions, and Appendix Table B.5 collects their coefficient on girls and how much of the gender gap each measure accounts for. Appendix Table B.6 reports the robustness checks; Appendix Tables B.7 and B.8, the bounds and the reweighting for selection into university; and Appendix Table B.9, the shares with the slopes corrected for assumed reliabilities of the test scores. Appendix Table B.10 reports the decomposition of the university gap by *Batxillerat* track; Appendix Table B.11, the slopes by gender and the gap by quartile of the 10th-grade math score; and Appendix Tables B.12 to B.15, the models with peer rank. Unless stated otherwise, the models include the controls and 10th-grade school and cohort fixed effects, with standard errors clustered by 10th-grade school.

	Science track							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.170***	−0.166***	−0.130***	−0.173***	−0.142***	−0.0972***	−0.194***	−0.0853***
	(0.00322)	(0.00315)	(0.00312)	(0.00309)	(0.00319)	(0.00302)	(0.00304)	(0.00327)
6th grade Math test score			0.140***					
			(0.00166)					
6th grade Reading test score				0.0902***				
				(0.00183)				
6th grade Math comp. adv.					0.0689***			
					(0.00174)			
10th grade Math test score						0.210***		
						(0.00169)		
10th grade Reading test score							0.114***	
							(0.00213)	
10th grade Math comp. adv.								0.141***
								(0.00186)
Constant	0.575***	0.487***	0.461***	0.483***	0.478***	0.443***	0.487***	0.458***
	(0.00269)	(0.00547)	(0.00535)	(0.00537)	(0.00548)	(0.00517)	(0.00535)	(0.00539)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	137,134	137,128	137,128	137,128	137,128	137,128	137,128	137,128
R-squared	0.029	0.058	0.111	0.078	0.069	0.177	0.086	0.104

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.1: **Science-track enrollment: gender gap and academic performance.** Linear probability models of enrollment in the Science track. Column 1 includes only the indicator for girls; column 2 adds the controls and 10th-grade school and cohort fixed effects; columns 3 to 8 add, one at a time, the 6th- and 10th-grade math score, reading score, and math comparative advantage, all in standard deviations. Controls: foreign-born student, two-parent household, parents of the same gender, all or one parent with a university degree, and one or both parents foreign-born. The coefficient on girls is the gender gap in the probability of enrollment. Students observed in *Batxillerat* with scores in both grades (the 2000–2006 birth cohorts; Section 2.1); columns 2 to 8 exclude 6 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	STEM degree							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.292***	−0.292***	−0.270***	−0.293***	−0.275***	−0.249***	−0.300***	−0.238***
	(0.00440)	(0.00442)	(0.00440)	(0.00440)	(0.00450)	(0.00439)	(0.00446)	(0.00458)
6th grade Math test score			0.0869***					
			(0.00236)					
6th grade Reading test score				0.0354***				
				(0.00276)				
6th grade Math comp. adv.					0.0598***			
					(0.00267)			
10th grade Math test score						0.134***		
						(0.00241)		
10th grade Reading test score							0.0463***	
							(0.00299)	
10th grade Math comp. adv.								0.109***
								(0.00269)
Constant	0.413***	0.378***	0.341***	0.366***	0.373***	0.305***	0.364***	0.352***
	(0.00403)	(0.00733)	(0.00725)	(0.00737)	(0.00727)	(0.00714)	(0.00729)	(0.00724)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	45,606	45,593	45,593	45,593	45,593	45,593	45,593	45,593
R-squared	0.113	0.143	0.164	0.146	0.153	0.193	0.148	0.174

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.2: **STEM-degree enrollment: gender gap and academic performance.** Linear probability models of enrollment in a STEM degree program under the main definition. Columns as in Appendix Table B.1. University entrants of the 2000–2003 birth cohorts observed in *Batxillerat*; columns 2 to 8 exclude 13 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	STEM degree (including life sciences and medicine)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.189***	−0.186***	−0.153***	−0.188***	−0.167***	−0.120***	−0.205***	−0.115***
	(0.00484)	(0.00480)	(0.00480)	(0.00475)	(0.00493)	(0.00472)	(0.00478)	(0.00505)
6th grade Math test score			0.130***					
			(0.00280)					
6th grade Reading test score				0.0765***				
				(0.00328)				
6th grade Math comp. adv.					0.0674***			
					(0.00308)			
10th grade Math test score						0.207***		
						(0.00283)		
10th grade Reading test score							0.104***	
							(0.00346)	
10th grade Math comp. adv.								0.142***
								(0.00307)
Constant	0.503***	0.429***	0.374***	0.403***	0.424***	0.316***	0.397***	0.396***
	(0.00408)	(0.00865)	(0.00856)	(0.00870)	(0.00857)	(0.00825)	(0.00853)	(0.00848)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	45,606	45,593	45,593	45,593	45,593	45,593	45,593	45,593
R-squared	0.037	0.073	0.109	0.084	0.082	0.166	0.091	0.113

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.3: **STEM-degree enrollment, definition including life sciences and medicine: gender gap and academic performance.** As Appendix Table B.2, for enrollment in a STEM degree program under the definition that also includes life sciences and medicine (Appendix Table A.1). The column label "STEM degree (including life sciences and medicine)" refers to this definition. University entrants of the 2000–2003 birth cohorts observed in *Batxillerat*; columns 2 to 8 exclude 13 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	STEM degree (math-intensive)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.286***	−0.287***	−0.267***	−0.288***	−0.272***	−0.248***	−0.295***	−0.239***
	(0.00424)	(0.00424)	(0.00423)	(0.00423)	(0.00433)	(0.00422)	(0.00426)	(0.00437)
6th grade Math test score			0.0791***					
			(0.00232)					
6th grade Reading test score				0.0347***				
				(0.00270)				
6th grade Math comp. adv.					0.0521***			
					(0.00260)			
10th grade Math test score						0.122***		
						(0.00238)		
10th grade Reading test score							0.0463***	
							(0.00295)	
10th grade Math comp. adv.								0.0960***
								(0.00258)
Constant	0.382***	0.355***	0.321***	0.343***	0.350***	0.288***	0.340***	0.332***
	(0.00396)	(0.00698)	(0.00691)	(0.00705)	(0.00691)	(0.00685)	(0.00699)	(0.00689)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	45,606	45,593	45,593	45,593	45,593	45,593	45,593	45,593
R-squared	0.118	0.147	0.166	0.151	0.155	0.193	0.152	0.173

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.4: **STEM-degree enrollment, math-intensive definition: gender gap and academic performance.** As Appendix Table B.2, for enrollment in a STEM degree program under the math-intensive definition (Appendix Table A.1). University entrants of the 2000–2003 birth cohorts observed in *Batxillerat*; columns 2 to 8 exclude 13 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	Outcome			
	Science track	**STEM degree**	**STEM degree (including life sciences and medicine)**	**STEM degree (math-intensive)**
Panel A. Coefficient on girls				
Mean of the outcome, boys	0.575	0.413	0.503	0.382
No controls or fixed effects	−0.170***	−0.292***	−0.189***	−0.286***
	(0.00322)	(0.00440)	(0.00484)	(0.00424)
With controls and fixed effects, no measure, α_0	−0.166***	−0.292***	−0.186***	−0.287***
	(0.00315)	(0.00442)	(0.00480)	(0.00424)
With 6th grade Math test score	−0.130***	−0.270***	−0.153***	−0.267***
	(0.00312)	(0.00440)	(0.00480)	(0.00423)
With 6th grade Reading test score	−0.173***	−0.293***	−0.188***	−0.288***
	(0.00309)	(0.00440)	(0.00475)	(0.00423)
With 6th grade Math comp. adv.	−0.142***	−0.275***	−0.167***	−0.272***
	(0.00319)	(0.00450)	(0.00493)	(0.00433)
With 10th grade Math test score	−0.0972***	−0.249***	−0.120***	−0.248***
	(0.00302)	(0.00439)	(0.00472)	(0.00422)
With 10th grade Reading test score	−0.194***	−0.300***	−0.205***	−0.295***
	(0.00304)	(0.00446)	(0.00478)	(0.00426)
With 10th grade Math comp. adv.	−0.0853***	−0.238***	−0.115***	−0.239***
	(0.00327)	(0.00458)	(0.00505)	(0.00437)
Panel B. Slope of the measure, β_A				
6th grade Math test score	0.140***	0.0869***	0.130***	0.0791***
	(0.00166)	(0.00236)	(0.00280)	(0.00232)
6th grade Reading test score	0.0902***	0.0354***	0.0765***	0.0347***
	(0.00183)	(0.00276)	(0.00328)	(0.00270)
6th grade Math comp. adv.	0.0689***	0.0598***	0.0674***	0.0521***
	(0.00174)	(0.00267)	(0.00308)	(0.00260)
10th grade Math test score	0.210***	0.134***	0.207***	0.122***
	(0.00169)	(0.00241)	(0.00283)	(0.00238)
10th grade Reading test score	0.114***	0.0463***	0.104***	0.0463***
	(0.00213)	(0.00299)	(0.00346)	(0.00295)
10th grade Math comp. adv.	0.141***	0.109***	0.142***	0.0960***
	(0.00186)	(0.00269)	(0.00307)	(0.00258)
Panel C. Conditional gender difference in the measure, θ_A, girls minus boys				
6th grade Math test score	−0.258	−0.258	−0.258	−0.258
6th grade Reading test score	0.0804	0.0260	0.0260	0.0260
6th grade Math comp. adv.	−0.339	−0.284	−0.284	−0.284
10th grade Math test score	−0.326	−0.321	−0.321	−0.321
10th grade Reading test score	0.245	0.178	0.178	0.178
10th grade Math comp. adv.	−0.571	−0.499	−0.499	−0.499
Panel D. Share of the gap accounted for (%)				
6th grade Math test score	21.8	7.7	18.0	7.1
6th grade Reading test score	−4.4	−0.3	−1.1	−0.3
6th grade Math comp. adv.	14.1	5.8	10.3	5.2
10th grade Math test score	41.3	14.7	35.7	13.6
10th grade Reading test score	−16.8	−2.8	−9.9	−2.9
10th grade Math comp. adv.	48.5	18.5	38.0	16.7
95 percent interval	[46.6, 50.8]	[17.4, 19.7]	[35.5, 40.8]	[15.7, 17.8]
Panel E. Math and comp. adv. entered jointly (10th grade)				
Coefficient on girls	−0.0911	−0.238	−0.115	−0.239
Share of the gap accounted for (%)	45.0	18.6	38.3	16.8
attributed to the Math test score (%)	88.2	66.0	89.1	70.2
attributed to Math comp. adv. (%)	11.8	34.0	10.9	29.8
95 percent interval	[8.9, 14.8]	[28.3, 38.8]	[5.9, 16.0]	[24.0, 35.0]
Same model, Math and Reading as the pair:				
attributed to the Math test score (%)	94.9	87.8	96.1	89.4
attributed to the Reading test score (%)	5.1	12.2	3.9	10.6
Math comp. adv. beyond Math, share of the gap (%)	5.3	6.3	4.2	5.0
Controls	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes
Observations	137,128	45,593	45,593	45,593
Observations, first two rows	137,134	45,606	45,606	45,606

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.5: **Accounting for the gender gap in STEM enrollment.** Linear probability models of enrollment in the Science track, among students observed in *Batxillerat* with scores in both grades (the 2000–2006 birth cohorts; Section 2.1), and in a STEM degree program under each definition, among university entrants of the 2000–2003 birth cohorts observed in *Batxillerat*. Panel A: the first two rows are the constant (boys' mean) and the coefficient on girls of a model without controls or fixed effects; the remaining rows add the controls of Appendix Table B.1 and 10th-grade school and cohort fixed effects, drop singleton fixed-effect groups (6 and 13 students), and report the coefficient on girls without a performance measure, α_0, and with one measure at a time, α_A. Panel B: slope of each measure, β_A, in standard deviations. Panel C: $\theta_A = (\alpha_0 - \alpha_A)/\beta_A$, which by the accounting identity is the coefficient on girls in a regression of the measure on girls, the controls, and the fixed effects; it does not depend on the outcome, so it is the same for the three STEM definitions. Panel D: share of α_0 accounted for, $100\,\beta_A\theta_A/\alpha_0$, negative when the gap widens. Panel E: the math score and math comparative advantage entered together, and the decomposition of Gelbach (2016), order-invariant but specific to this pair (Section 3), and the same model split with the math and reading scores as the pair; its last row is the comparative-advantage contribution as a share of α_0. The 95 percent intervals are the 2.5th and 97.5th percentiles of cluster-bootstrap replications that resample 10th-grade schools (999 replications in Panel D, 500 in Panel E). Panels C to E use unrounded coefficients, so recomputing them from Panels A and B can move the last digit. Math comp. adv.: standardized math minus standardized reading score. Appendix Tables B.1 to B.4 report the full models, and Appendix Table B.6 the robustness checks. Source: Idescat administrative records; authors' calculations.

	Outcome			
	Science track	**STEM degree**	**STEM degree (including life sciences and medicine)**	**STEM degree (math-intensive)**
Panel A. Share of the gap accounted for by 10th grade Math comp. adv. (%)				
Main specification	48.5	18.5	38.0	16.7
School and cohort FE, common sample	48.5	18.5	38.0	16.6
School × cohort FE, common sample	50.4	19.3	39.3	17.4
Reading as the higher language score	45.7	17.2	35.8	15.5
Math comp. adv. as z_{M-R}	61.4	21.3	47.6	19.4
Math comp. adv. as a percentile-rank difference	43.9	17.4	34.6	15.7
Panel B. Girls × 10th grade Math comp. adv.				
Math comp. adv. as $z_M - z_R$	0.00379	-0.0657^{***}	-0.00940	-0.0749^{***}
	(0.00338)	(0.00552)	(0.00608)	(0.00547)
Math comp. adv. as z_{M-R}	-0.0214^{***}	-0.0975^{***}	-0.0417^{***}	-0.105^{***}
	(0.00266)	(0.00459)	(0.00505)	(0.00454)
Panel C. Math comp. adv. beyond the 10th grade Math test score, share of the gap (%)				
Main specification	5.3	6.3	4.2	5.0
Observations, main specification	137,128	45,593	45,593	45,593

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.6: **Robustness of the main results.** Panel A: share of the gender gap in enrollment accounted for by 10th-grade math comparative advantage, $100(\alpha_0 - \alpha_A)/\alpha_0$, in percent. The first row is the main specification; each other row changes one element. The next two rows use additive school and cohort fixed effects and school-by-cohort fixed effects on the students common to the two structures (137,057 in the Science-track sample and 45,420 in the university sample); the first of them is the benchmark for the second. The last three measure reading as the higher of the Catalan and Spanish scores, comparative advantage as z_{M-R}, and comparative advantage as the difference between the percentile ranks of the two scores within the cohort, which does not depend on how either score is scaled. Panel B: interaction between the indicator for girls and 10th-grade math comparative advantage under each construction, from models as in Appendix Table B.11, Panel A. Panel C: the part of the gap that math comparative advantage accounts for beyond the 10th-grade math score, as a share of α_0, from the decomposition of Appendix Table B.5, Panel E. Clustering by both the 6th- and the 10th-grade school instead of the 10th-grade school alone, on the students with an observed 6th-grade school (137,116 and 45,584), leaves the point estimates identical and moves the standard error of the coefficient on girls, in the model with comparative advantage, by at most 0.0001 in any column. Samples and controls as in Appendix Table B.5; the two additional definitions of a STEM degree use its university sample. Source: Idescat administrative records; authors' calculations.

	Gender gap among university entrants, girls − boys (percentage points)							
	Students observed in *Batxillerat*				All students of the 2000–2003 cohorts			
	Raw gap	Lower bound	Upper bound	Entrants	Raw gap	Lower bound	Upper bound	Entrants
Panel A. STEM degree								
All entrants	−29.2	−37.8	−28.1	45,606	−29.1	−41.0	−25.5	46,575
		(0.72)	(0.43)			(0.35)	(0.45)	
10th grade Math quartile 1	−10.4	−12.9	−9.1	3,697	−10.8	−13.1	−7.4	3,967
		(1.18)	(1.27)			(1.13)	(1.33)	
10th grade Math quartile 2	−15.2	−20.7	−12.8	8,791	−15.4	−20.9	−10.1	9,095
		(0.78)	(0.89)			(0.76)	(0.97)	
10th grade Math quartile 3	−23.2	−33.8	−20.7	13,773	−23.2	−33.7	−18.0	14,003
		(0.63)	(0.77)			(0.62)	(0.82)	
10th grade Math quartile 4	−32.5	−39.1	−30.8	19,345	−32.5	−44.6	−29.4	19,510
		(0.88)	(0.69)			(0.95)	(0.71)	
Trimming within math quartiles		−33.9	−24.5					
		(0.53)	(0.46)					
Quartile 4 minus quartile 1	−22.1	−17.9			−21.7	−16.3		
Panel B. STEM degree (math-intensive)								
All entrants	−28.6	−37.5	−27.7	45,606	−28.5	−37.9	−25.6	46,575
		(0.72)	(0.41)			(0.35)	(0.43)	
10th grade Math quartile 1	−9.7	−11.5	−8.8	3,697	−10.2	−11.9	−7.7	3,967
		(1.13)	(1.19)			(1.08)	(1.23)	
10th grade Math quartile 2	−14.5	−18.6	−12.7	8,791	−14.8	−18.9	−10.9	9,095
		(0.75)	(0.84)			(0.73)	(0.89)	
10th grade Math quartile 3	−22.8	−30.9	−20.9	13,773	−22.8	−30.9	−18.9	14,003
		(0.62)	(0.73)			(0.61)	(0.77)	
10th grade Math quartile 4	−32.7	−39.6	−31.3	19,345	−32.7	−45.4	−30.1	19,510
		(0.88)	(0.66)			(0.96)	(0.68)	
Trimming within math quartiles		−32.9	−24.7					
		(0.52)	(0.44)					
Quartile 4 minus quartile 1	−23.0	−19.8			−22.5	−18.3		
Panel C. STEM degree (including life sciences and medicine)								
All entrants	−18.9	−25.6	−15.8	45,606	−18.8	−39.7	−9.2	46,575
		(0.66)	(0.52)			(0.81)	(0.59)	
10th grade Math quartile 1	−5.3	−16.7	0.9	3,697	−5.8	−17.9	11.8	3,967
		(1.32)	(1.71)			(1.28)	(2.21)	
10th grade Math quartile 2	−9.5	−26.8	−2.2	8,791	−10.2	−27.8	6.5	9,095
		(0.85)	(1.17)			(0.84)	(1.41)	
10th grade Math quartile 3	−12.9	−29.7	−6.1	13,773	−12.8	−41.5	1.3	14,003
		(1.30)	(0.98)			(0.65)	(1.11)	
10th grade Math quartile 4	−14.7	−18.8	−10.6	19,345	−14.7	−22.4	−7.1	19,510
		(0.82)	(0.82)			(0.87)	(0.87)	
Trimming within math quartiles		−23.0	−7.6					
		(0.63)	(0.57)					
Quartile 4 minus quartile 1	−9.3	6.2			−8.9	10.8		

Standard errors of the bounds in parentheses: analytic (Lee 2009), not clustered
Entering university: 62.9 percent of girls and 57.3 percent of boys observed in *Batxillerat*; 50.6 and 38.8 percent of all students

Table B.7: **STEM-degree gap and selection into university: Lee bounds.** Gap in STEM-degree enrollment between girls and boys among university entrants of the 2000–2003 birth cohorts, overall and by quartile of the 10th-grade math score, without controls or fixed effects, by STEM definition. Lower and upper bounds: bounds of Lee (2009), which trim the excess share of girls among entrants from either end of girls' outcomes; they assume that no student who would enter university as a boy would fail to enter as a girl and that gender is otherwise independent of the outcome. Left block: students observed in *Batxillerat*. Right block: all students of the analysis sample in those cohorts; a student who enters university without being observed in *Batxillerat* counts as an entrant. Trimming within math quartiles: the trimming done separately in each quartile, a description of the data rather than a sharper bound, since the quartile is not independent of gender. Quartile 4 minus quartile 1: the raw difference, and the most adverse combination of the within-quartile bounds, the upper bound in quartile 4 minus the lower bound in quartile 1 – descriptive rather than a bound on the difference, since the quartile is not independent of gender. Source: Idescat administrative records; authors' calculations.

	STEM degree		STEM degree (math-intensive)		STEM degree (including life sciences and medicine)	
	Unweighted	Reweighted	Unweighted	Reweighted	Unweighted	Reweighted
Coefficient on girls, no performance measure	-0.292^{***}	-0.274^{***}	-0.287^{***}	-0.269^{***}	-0.186^{***}	-0.182^{***}
	(0.004)	(0.005)	(0.004)	(0.005)	(0.005)	(0.006)
Coefficient on girls, with 10th grade Math comp. adv.	-0.238^{***}	-0.223^{***}	-0.239^{***}	-0.222^{***}	-0.115^{***}	-0.115^{***}
	(0.005)	(0.006)	(0.004)	(0.005)	(0.005)	(0.006)
Share of the gap accounted for (%)	18.5	18.9	16.7	17.3	38.0	37.0
Controls	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	45,593	45,593	45,593	45,593	45,593	45,593

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.8: **STEM-degree gap and selection into university: reweighting.** Linear probability models of STEM-degree enrollment among university entrants of the 2000–2003 birth cohorts observed in *Batxillerat*, with the controls of Appendix Table B.1 and 10th-grade school and cohort fixed effects, without a performance measure and with 10th-grade math comparative advantage, as in Appendix Table B.5. Reweighted: entrants weighted by the inverse of their probability of entering university, from a logit of entry on gender, the 10th-grade math score and comparative advantage, the controls, and cohort, estimated on the students observed in *Batxillerat*; the weights correct only for selection on these observed characteristics. The share is $100(\alpha_0 - \alpha_A)/\alpha_0$ under each weighting. Source: Idescat administrative records; authors' calculations.

Reliability		Error corr.	Science track				STEM degree			
Math	Reading	ρ	Math, alone (%)	Math comp. adv., alone (%)	Gelbach: Math comp. adv. part (%)	Beyond Math, share of gap (%)	Math, alone (%)	Math comp. adv., alone (%)	Gelbach: Math comp. adv. part (%)	Beyond Math, share of gap (%)
Panel A. Independent errors										
1.00	1.00	0.0	41.3	48.5	11.8	5.3	14.7	18.5	34.0	6.3
0.95	0.95	0.0	44.6	58.7	22.5	12.0	16.2	23.4	44.1	10.0
0.90	0.90	0.0	48.4	74.3	35.3	23.2	18.1	31.7	57.2	17.0
0.90	0.80	0.0	48.4	[101.2][a]	49.2	37.6	18.1	49.1	85.8	37.9
0.80	0.90	0.0	58.6	[101.2][a]	51.8	52.4	23.6	49.1	67.4	34.0
Panel B. Positively correlated Math and Reading errors										
0.95	0.95	0.2	44.6	56.3	18.2	9.3	16.2	22.2	39.9	8.7
0.90	0.90	0.2	48.4	67.2	25.9	15.4	18.1	27.7	47.7	12.6
0.85	0.85	0.2	53.0	83.1	35.4	25.5	20.5	36.9	58.4	19.9
0.95	0.95	0.4	44.6	54.2	13.7	6.8	16.2	21.2	35.7	7.4
0.90	0.90	0.4	48.4	61.3	16.1	8.8	18.1	24.7	37.9	9.0
0.85	0.85	0.4	53.0	70.5	19.1	11.7	20.5	29.6	41.1	11.3
Panel C. Reading reliability from the Catalan and Spanish scores (Spearman-Brown, 0.82)										
1.00	0.82	0.0	41.3	71.1	20.8	10.1	14.7	29.9	64.8	16.0
0.90	0.82	0.0	48.4	95.4	46.2	34.1	18.1	44.9	79.2	31.5
0.82	0.82	0.0	56.6	[133.3][a]	63.0	71.8	22.4	77.4	89.7	64.9
Breakdown reliability, joint model					0.72				0.77	
Breakdown reliability, Math comp. adv. alone					0.71				0.76	
Math comp. adv. alone reaches 100 percent at					0.85				0.80	
Controls					Yes				Yes	
School and cohort FE					Yes				Yes	
Observations					137,128				45,593	

Table B.9: **Gender gaps in STEM enrollment: the shares under assumed reliabilities of the test scores.** Shares of the conditional gender gap, in percent, with the slopes of the 10th-grade math and reading scores corrected for classical measurement error. Models and samples as in Appendix Table B.5; the first row, with both reliabilities at 1.00, reproduces its Panels C and D. "Beyond Math" is the comparative-advantage part of the joint model as a share of the gap left after the math score. Reliability is the assumed share of each score's within-cohort variance that is not error. The errors are uncorrelated except in Panel B, where they correlate with ρ; Panel C sets reading at 0.82 (Spearman-Brown, from the Catalan and Spanish scores). Figures in brackets, marked a, exceed 100 percent and are not shares. Breakdown reliability: the value below which the correction is not defined. This is a sensitivity calculation, so no standard errors are reported. Source: Idescat administrative records; authors' calculations.

	STEM degree (main)	STEM degree (math-intensive)	STEM degree (including life sciences and medicine)
Panel A. The gap and its three terms (percentage points)			
Gap, boys minus girls	29.25***	28.64***	18.91***
	(0.44)	(0.42)	(0.48)
Track composition, at the STEM rates of girls	4.69***	3.79***	12.26***
	(0.16)	(0.13)	(0.35)
STEM rates within tracks, at the track shares of girls	17.94***	18.12***	5.25***
	(0.34)	(0.33)	(0.31)
Interaction	6.61***	6.74***	1.40***
	(0.24)	(0.24)	(0.15)
Panel B. Share of the gap due to track composition (percent)			
STEM rates of girls as reference	16.04***	13.22***	64.83***
	(0.52)	(0.46)	(1.49)
STEM rates of boys as reference	38.65***	36.75***	72.26***
	(0.91)	(0.89)	(1.36)
Average of the two (Kitagawa)	27.35***	24.99***	68.54***
	(0.68)	(0.64)	(1.38)
Panel C. Kitagawa within-track term, by track (percentage points)			
Within-track term, total	21.25***	21.49***	5.95***
	(0.37)	(0.36)	(0.33)
Science track	20.28***	20.59***	5.14***
	(0.35)	(0.34)	(0.31)
Humanities and social sciences track	0.83***	0.74***	0.69***
	(0.10)	(0.10)	(0.12)
Arts or general track	0.14	0.15*	0.13
	(0.07)	(0.07)	(0.08)
Observations	45,606	45,606	45,606
10th-grade schools	1,074	1,074	1,074
Controls	No	No	No
School and cohort FE	No	No	No

Boys minus girls, raw means. Delta-method standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.10: **Where the STEM-degree gap arises: *Batxillerat* track composition or choice within the track.** Raw decomposition of the gender gap in enrollment in a STEM degree program among university entrants of the 2000–2003 birth cohorts, by STEM definition, in percentage points (boys minus girls). The share who enter a STEM degree program is an average of the rates within each track, weighted by the share of entrants in that track, so the gap splits into the difference in track shares (composition), the difference in the rates within tracks, and their interaction. Panel B reports the composition part as a share of the gap under three ways of allocating the interaction: valuing the difference in track shares at girls' rates, at boys' rates, and at the average of the two (Kitagawa, 1955). Panel C splits the last of these within-track terms by track. Tracks are Science, Humanities and social sciences, and Arts or general, the last two grouped as few university entrants come from either. Standard errors, clustered by 10th-grade school, are obtained by the delta method from the joint estimation of the underlying means. Source: Idescat administrative records; authors' calculations.

	Outcome	
	Science track	**STEM degree**
Panel A. Slope of the measure, by gender		
6th grade Math test score		
Boys	0.154***	0.137***
	(0.00260)	(0.00503)
Girls	0.131***	0.0604***
	(0.00192)	(0.00236)
Girls − boys	−0.0236***	−0.0769***
	(0.00300)	(0.00551)
6th grade Reading test score		
Boys	0.0878***	0.0481***
	(0.00263)	(0.00526)
Girls	0.0921***	0.0267***
	(0.00217)	(0.00305)
Girls − boys	0.00425	−0.0213***
	(0.00308)	(0.00612)
6th grade Math comp. adv.		
Boys	0.0664***	0.0833***
	(0.00267)	(0.00525)
Girls	0.0708***	0.0447***
	(0.00223)	(0.00273)
Girls − boys	0.00434	−0.0386***
	(0.00341)	(0.00592)
10th grade Math test score		
Boys	0.229***	0.214***
	(0.00238)	(0.00479)
Girls	0.197***	0.0916***
	(0.00199)	(0.00246)
Girls − boys	−0.0314***	−0.122***
	(0.00272)	(0.00524)
10th grade Reading test score		
Boys	0.105***	0.0580***
	(0.00268)	(0.00522)
Girls	0.122***	0.0363***
	(0.00283)	(0.00334)
Girls − boys	0.0164***	−0.0217***
	(0.00348)	(0.00621)
10th grade Math comp. adv.		
Boys	0.139***	0.148***
	(0.00264)	(0.00504)
Girls	0.142***	0.0819***
	(0.00239)	(0.00264)
Girls − boys	0.00379	−0.0657***
	(0.00338)	(0.00552)
Panel B. Gender gap within each quartile of the 10th grade math score		
Quartile 1 (lowest)	−0.0762***	−0.102***
	(0.00583)	(0.0130)
Quartile 2	−0.0906***	−0.148***
	(0.00522)	(0.00882)
Quartile 3	−0.103***	−0.232***
	(0.00493)	(0.00728)
Quartile 4 (highest)	−0.108***	−0.326***
	(0.00421)	(0.00669)
p-value: equal gender gap across quartiles	<0.001	<0.001
Observations, Panel B	155,486	45,593
Controls	Yes	Yes
School and cohort FE	Yes	Yes
Observations, Panel A	137,128	45,593

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.11: **Slopes by gender and the gender gap by math quartile.** Samples, controls, fixed effects, and clustering as in Appendix Table B.5, except that Panel B, which uses no 6th-grade measure, also includes the 2007 birth cohort in the Science track. Panel A: one model per performance measure, interacting the measure with the indicator for girls. The row "Boys" is the slope of the measure, the row "Girls" the slope for girls, the sum of the two coefficients, with the standard error of that linear combination, and the row "Girls − boys" the interaction. Panel B: level of the conditional gender gap within each quartile of the 10th-grade math score, from a single model interacting the quartile with the indicator for girls. The gap in quartile 1 is the coefficient on girls, and each of the others is that coefficient plus the corresponding interaction, with its standard error computed for the linear combination; the last row tests that the four gaps are equal. Quartiles are defined in the analysis sample, so university entrants are concentrated in the upper quartiles. Source: Idescat administrative records; authors' calculations.

	Science track							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.164***	−0.159***	−0.0856***	−0.189***	−0.122***	−0.0727***	−0.0884***	−0.141***
	(0.00470)	(0.00451)	(0.00431)	(0.00423)	(0.00428)	(0.00466)	(0.00436)	(0.00440)
10th grade Math test score			0.214***					
			(0.00239)					
10th grade Reading test score				0.116***				
				(0.00279)				
10th grade Science test score					0.198***			
					(0.00222)			
10th grade Math comp. adv.						0.145***		
						(0.00250)		
10th grade Math peer rank (standardized)							0.193***	
							(0.00207)	
10th grade Math same-gender peer rank (standardized)								0.185***
								(0.00200)
Constant	0.578***	0.478***	0.433***	0.478***	0.447***	0.448***	0.447***	0.474***
	(0.00385)	(0.00820)	(0.00746)	(0.00799)	(0.00774)	(0.00791)	(0.00750)	(0.00759)
Share of the gap accounted for (percent)			46.2	−18.6	23.3	54.4	44.5	11.6
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	68,095	68,095	68,095	68,095	68,095	68,095	68,095	68,095
R-squared	0.027	0.058	0.182	0.086	0.167	0.106	0.182	0.174

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.12: **Science-track enrollment: gender gap, academic performance, and peer rank.** As Appendix Table B.1, adding one at a time the 10th-grade math score, reading score, science score, math comparative advantage, standardized peer rank in math, and standardized rank among same-gender schoolmates (columns 3 to 8). Sample: students observed in *Batxillerat* with a 10th-grade science score, from the 2001–2007 birth cohorts, whose school-cohort cell holds more than 32 students in the data and at least 10 students of their own gender (N = 68,095). The 2000 birth cohort is left out because the extract lacks its 10th-grade science scores, from 2015–16. Scores are standardized within cohort on this sample, so one standard deviation here is not one standard deviation of the analysis sample. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	STEM degree							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.296***	−0.296***	−0.254***	−0.306***	−0.270***	−0.240***	−0.254***	−0.280***
	(0.00724)	(0.00731)	(0.00726)	(0.00739)	(0.00708)	(0.00777)	(0.00723)	(0.00722)
10th grade Math test score			0.137***					
			(0.00400)					
10th grade Reading test score				0.0473***				
				(0.00509)				
10th grade Science test score					0.119***			
					(0.00381)			
10th grade Math comp. adv.						0.111***		
						(0.00457)		
10th grade Math peer rank (standardized)							0.117***	
							(0.00327)	
10th grade Math same-gender peer rank (standardized)								0.115***
								(0.00327)
Constant	0.426***	0.395***	0.324***	0.381***	0.339***	0.370***	0.343***	0.358***
	(0.00663)	(0.0126)	(0.0121)	(0.0126)	(0.0122)	(0.0122)	(0.0120)	(0.0121)
Share of the gap accounted for (percent)			14.4	−3.2	8.7	19.0	14.2	5.5
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,691	15,691	15,691	15,691	15,691	15,691	15,691	15,691
R-squared	0.113	0.147	0.197	0.151	0.188	0.177	0.199	0.198

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.13: **STEM-degree enrollment: gender gap, academic performance, and peer rank.** As Appendix Table B.12, for enrollment in a STEM degree program under the main definition. Sample: university entrants observed in *Batxillerat* with a 10th-grade science score, from the 2001–2003 birth cohorts, whose school-cohort cell holds more than 32 students in the data and at least 10 students of their own gender (N = 15,691). The science criterion leaves out the 2000 birth cohort, so three of the four birth cohorts that enter university in the years observed remain. Scores are standardized within cohort on the science sample, so one standard deviation here is not one standard deviation of the analysis sample. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	Science track								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Girls	−0.160***	−0.0872***	−0.0898***	−0.0865***	−0.0852***	−0.0751***	−0.0792***	−0.0802***	−0.102***
	(0.00431)	(0.00412)	(0.00414)	(0.00412)	(0.00411)	(0.00447)	(0.00419)	(0.00416)	(0.00433)
10th grade Math test score		0.216***		0.113***	0.145***			0.105***	0.155***
		(0.00229)		(0.00618)	(0.00745)			(0.00624)	(0.00528)
10th grade Math comp. adv.						0.145***	0.0253***	0.0155***	
						(0.00240)	(0.00279)	(0.00282)	
10th grade Math peer rank (standardized)			0.194***	0.0977***	0.0575***		0.182***	0.0977***	
			(0.00193)	(0.00561)	(0.00604)		(0.00237)	(0.00560)	
10th grade Math same-gender peer rank (standardized)									0.0588***
									(0.00462)
Constant	0.481***	0.434***	0.450***	0.441***	0.405***	0.451***	0.447***	0.440***	0.446***
	(0.00760)	(0.00703)	(0.00705)	(0.00703)	(0.00718)	(0.00742)	(0.00707)	(0.00704)	(0.00706)
Share of the gap accounted for (%)	0.0	45.4	43.8	45.8	46.7	53.0	50.4	49.7	36.0
Part of the gap attributed to the Math test score (%)				23.8					
Part of the gap attributed to peer rank (%)				22.1					
Cubic in Math test score	No	No	No	No	Yes	No	No	No	No
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	78,325	78,325	78,325	78,325	78,325	78,325	78,325	78,325	78,325
R-squared	0.057	0.181	0.181	0.184	0.189	0.105	0.182	0.184	0.183

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.14: **Science-track enrollment: gender gap, math score, and peer rank.** Linear probability models with the controls of Appendix Table B.1 and 10th-grade school and cohort fixed effects: (1) no measure of academic performance; (2) the 10th-grade math score; (3) peer rank in math; (4) both; (5) a cubic in the math score, plus peer rank (the squared and cubed terms are not shown); (6) math comparative advantage; (7) math comparative advantage and peer rank; (8) the math score, math comparative advantage, and peer rank; (9) the math score and the rank among same-gender schoolmates. Below column 4, the decomposition of Gelbach (2016) divides its reduction of the gap between the math score and peer rank, each as a percentage of the gap in column 1. Scores and ranks in standard deviations. Students observed in *Batxillerat* from the 2000–2007 birth cohorts, in school-cohort cells with more than 32 students in the data and with at least 10 students of their own gender (N = 78,325). No science score is required here, so this sample is wider than that of Appendix Table B.12. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	STEM degree								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Girls	−0.292***	−0.249***	−0.250***	−0.248***	−0.247***	−0.236***	−0.235***	−0.235***	−0.260***
	(0.00606)	(0.00597)	(0.00591)	(0.00595)	(0.00594)	(0.00625)	(0.00619)	(0.00619)	(0.00626)
10th grade Math test score		0.138***		0.0596***	0.0644***			0.0325**	0.0780***
		(0.00356)		(0.0107)	(0.0123)			(0.0110)	(0.00866)
10th grade Math comp. adv.						0.112***	0.0451***	0.0413***	
						(0.00400)	(0.00496)	(0.00510)	
10th grade Math peer rank (standardized)			0.117***	0.0698***	0.0313**		0.0965***	0.0725***	
			(0.00294)	(0.00913)	(0.0100)		(0.00369)	(0.00915)	
10th grade Math same-gender peer rank (standardized)									0.0545***
									(0.00716)
Constant	0.386***	0.313***	0.336***	0.325***	0.291***	0.361***	0.335***	0.329***	0.329***
	(0.0104)	(0.0102)	(0.0100)	(0.0103)	(0.0109)	(0.0103)	(0.0100)	(0.0102)	(0.0104)
Share of the gap accounted for (%)	−0.0	14.8	14.5	15.1	15.6	19.2	19.7	19.5	10.9
Part of the gap attributed to the Math test score (%)				6.4					
Part of the gap attributed to peer rank (%)				8.7					
Cubic in Math test score	No	No	No	No	Yes	No	No	No	No
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,727	21,727	21,727	21,727	21,727	21,727	21,727	21,727	21,727
R-squared	0.140	0.192	0.193	0.194	0.200	0.172	0.196	0.197	0.194

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.15: **STEM-degree enrollment: gender gap, math score, and peer rank.** As Appendix Table B.14, for enrollment in a STEM degree program under the main definition. University entrants of the 2000–2003 birth cohorts observed in *Batxillerat*, in school-cohort cells with more than 32 students in the data and with at least 10 students of their own gender (N = 21,727). No science score is required here, so this sample is wider than that of Appendix Table B.13. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

Appendix C. Results in the Analysis Sample, Without Conditioning on Post-Compulsory Enrollment

This appendix reports the models of Section 4 in the analysis sample, where the outcomes do not condition on enrollment: a student who never enters *Batxillerat* counts as not enrolled in the Science track, and a student of the 2000–2003 birth cohorts who does not enter a Catalan university counts as not enrolled in a STEM degree. Appendix Table C.1 summarizes the models and Appendix Tables C.2 to C.5 report them in full. Each outcome combines two decisions, continuing beyond compulsory schooling and choosing STEM at the stage reached, so these estimates describe overall STEM participation rather than field choice among enrollees, which is why the main results condition on enrollment. Like the core analysis of Breda and Napp (2019), these models relate scores measured around age 15 to an outcome defined for all students, not only for those who continue. Their outcome, however, is a stated intention to pursue math-intensive studies and careers rather than realized enrollment, so the two sets of estimates are not directly comparable.

With controls and fixed effects, girls are 5.5 percentage points less likely than boys to enroll in the Science track, against 16.6 points among students observed in *Batxillerat*, because the two decisions work in opposite directions: more girls than boys continue to *Batxillerat*, and fewer of those who continue choose the Science track (Section 2.1). Adding the 10th-grade mathematics score reduces the gap to 0.8 points, 85.6% less, and adding math comparative advantage to 0.7 points, 86.6% less, while the reading score more than doubles it, to 11.1 points. In the 2000–2003 birth cohorts, girls are 9.7 percentage points less likely to enroll in a STEM degree program. The mathematics score reduces the gap by 18.3%, to 7.9 points, and comparative advantage by 20.3%, to 7.7 points, close to the 14.7% and 18.5% among university entrants (Section 4.1). Under the math-intensive definition the shares are 16.3% and 17.5%, and under the definition that also includes life sciences and medicine, on a gap of 3.3 points, 81.8% and 83.3% (Appendix Table C.1).

	Unconditional on post-compulsory education enrollment			
	Science track	**STEM degree (main)**	**STEM degree (including life sciences and medicine)**	**STEM degree (math-intensive)**
Mean of the outcome, boys	0.372	0.159	0.195	0.147
Gender gap, no controls or fixed effects	−0.0579***	−0.0987***	−0.0359***	−0.0995***
	(0.00285)	(0.00247)	(0.00253)	(0.00235)
Panel A. Coefficient on girls				
No measure, α_0	−0.0552***	−0.0970***	−0.0331***	−0.0980***
	(0.00255)	(0.00242)	(0.00240)	(0.00231)
With 6th grade Math test score	−0.0270***	−0.0855***	−0.0158***	−0.0878***
	(0.00247)	(0.00234)	(0.00231)	(0.00223)
With 6th grade Reading test score	−0.0790***	−0.105***	−0.0463***	−0.106***
	(0.00248)	(0.00244)	(0.00238)	(0.00234)
With 6th grade Math comp. adv.	−0.0439***	−0.0925***	−0.0281***	−0.0943***
	(0.00259)	(0.00239)	(0.00243)	(0.00229)
With 10th grade Math test score	−0.00793***	−0.0792***	−0.00603**	−0.0820***
	(0.00233)	(0.00227)	(0.00222)	(0.00217)
With 10th grade Reading test score	−0.111***	−0.119***	−0.0684***	−0.118***
	(0.00244)	(0.00254)	(0.00243)	(0.00242)
With 10th grade Math comp. adv.	−0.00739**	−0.0773***	−0.00553*	−0.0808***
	(0.00261)	(0.00233)	(0.00245)	(0.00222)
Panel B. Slope of the measure				
6th grade Math test score	0.142***	0.0597***	0.0903***	0.0537***
6th grade Reading test score	0.126***	0.0523***	0.0829***	0.0476***
6th grade Math comp. adv.	0.0290***	0.0127***	0.0145***	0.0107***
10th grade Math test score	0.205***	0.0872***	0.133***	0.0787***
10th grade Reading test score	0.145***	0.0612***	0.0967***	0.0559***
10th grade Math comp. adv.	0.0781***	0.0347***	0.0486***	0.0303***
Panel C. Conditional gender difference in the measure (girls minus boys)				
6th grade Math test score	−0.199	−0.191	−0.191	−0.191
6th grade Reading test score	0.190	0.159	0.159	0.159
6th grade Math comp. adv.	−0.388	−0.350	−0.350	−0.350
10th grade Math test score	−0.230	−0.203	−0.203	−0.203
10th grade Reading test score	0.382	0.364	0.364	0.364
10th grade Math comp. adv.	−0.612	−0.568	−0.568	−0.568
Panel D. Share of the gap accounted for (%)				
6th grade Math test score	51.1	11.8	52.2	10.5
6th grade Reading test score	−43.2	−8.6	−39.7	−7.7
6th grade Math comp. adv.	20.4	4.6	15.3	3.8
10th grade Math test score	85.6	18.3	81.8	16.3
10th grade Reading test score	−100.3	−23.0	−106.3	−20.8
10th grade Math comp. adv.	86.6	20.3	83.3	17.5
Controls	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes
Observations	192,312	103,948	103,948	103,948
Observations, first two rows	192,319	103,952	103,952	103,952

Standard errors, clustered by 10th-grade school, in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.1: **Accounting for the gender gap in STEM enrollment in the analysis sample.** Models as in Appendix Tables C.2 to C.5: the Science track in the analysis sample and the three STEM degree definitions in its 2000–2003 birth cohorts, with outcomes that do not condition on enrollment. Panel A: coefficient on girls without a performance measure, α_0, and with each measure, α_A. Panel B: slope of each measure, β_A, in standard deviations. Panel C: θ_A, the coefficient on girls in a regression of the measure on girls, the controls, and the fixed effects. Panel D: share of α_0 accounted for, $100(\alpha_0 - \alpha_A)/\alpha_0$, negative when the gap widens. The first two rows give the boys' mean and the raw gap. Panel B omits standard errors, which are in Appendix Tables C.2 to C.5. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	Unconditional on post-compulsory education enrollment							
	Science track							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.0579***	−0.0552***	−0.0270***	−0.0790***	−0.0439***	−0.00793***	−0.111***	−0.00739**
	(0.00285)	(0.00255)	(0.00247)	(0.00248)	(0.00259)	(0.00233)	(0.00244)	(0.00261)
6th grade Math test score			0.142***					
			(0.00115)					
6th grade Reading test score				0.126***				
				(0.00115)				
6th grade Math comp. adv.					0.0290***			
					(0.00131)			
10th grade Math test score						0.205***		
						(0.00125)		
10th grade Reading test score							0.145***	
							(0.00138)	
10th grade Math comp. adv.								0.0781***
								(0.00161)
Constant	0.372***	0.249***	0.269***	0.293***	0.243***	0.291***	0.317***	0.229***
	(0.00361)	(0.00409)	(0.00389)	(0.00386)	(0.00410)	(0.00359)	(0.00382)	(0.00407)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	192,319	192,312	192,312	192,312	192,312	192,312	192,312	192,312
R-squared	0.004	0.069	0.148	0.129	0.071	0.225	0.146	0.086

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.2: **Science-track enrollment in the analysis sample: gender gap and academic performance.** As Appendix Table B.1, in the analysis sample, where students never observed in *Batxillerat* count as not enrolled, restricted like that table to the students with scores in both grades (the 2000–2006 birth cohorts). Columns 2 to 8 exclude 7 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	Unconditional on post-compulsory education enrollment							
	STEM degree							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.0987***	−0.0970***	−0.0855***	−0.105***	−0.0925***	−0.0792***	−0.119***	−0.0773***
	(0.00247)	(0.00242)	(0.00234)	(0.00244)	(0.00239)	(0.00227)	(0.00254)	(0.00233)
6th grade Math test score			0.0597***					
			(0.000998)					
6th grade Reading test score				0.0523***				
				(0.000972)				
6th grade Math comp. adv.					0.0127***			
					(0.00100)			
10th grade Math test score						0.0872***		
						(0.00125)		
10th grade Reading test score							0.0612***	
							(0.00105)	
10th grade Math comp. adv.								0.0347***
								(0.00115)
Constant	0.159***	0.108***	0.114***	0.124***	0.105***	0.123***	0.134***	0.0992***
	(0.00285)	(0.00290)	(0.00290)	(0.00290)	(0.00290)	(0.00281)	(0.00292)	(0.00288)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	103,952	103,948	103,948	103,948	103,948	103,948	103,948	103,948
R-squared	0.025	0.072	0.105	0.096	0.073	0.138	0.104	0.080

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.3: **STEM-degree enrollment in the analysis sample, 2000–2003 birth cohorts: gender gap and academic performance.** As Appendix Table B.2, in the analysis sample of the 2000–2003 birth cohorts, where students who do not enter a Catalan university between 2018–19 and 2021–22 count as not enrolled and university entrants never observed in *Batxillerat* are classified by their degree program. Columns 2 to 8 exclude 4 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	Unconditional on post-compulsory education enrollment							
	STEM degree (including life sciences and medicine)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.0359***	−0.0331***	−0.0158***	−0.0463***	−0.0281***	−0.00603**	−0.0684***	−0.00553*
	(0.00253)	(0.00240)	(0.00231)	(0.00238)	(0.00243)	(0.00222)	(0.00243)	(0.00245)
6th grade Math test score			0.0903***					
			(0.00129)					
6th grade Reading test score				0.0829***				
				(0.00126)				
6th grade Math comp. adv.					0.0145***			
					(0.00122)			
10th grade Math test score						0.133***		
						(0.00160)		
10th grade Reading test score							0.0967***	
							(0.00143)	
10th grade Math comp. adv.								0.0486***
								(0.00142)
Constant	0.195***	0.110***	0.119***	0.135***	0.107***	0.133***	0.151***	0.0973***
	(0.00337)	(0.00356)	(0.00346)	(0.00346)	(0.00356)	(0.00328)	(0.00344)	(0.00356)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	103,952	103,948	103,948	103,948	103,948	103,948	103,948	103,948
R-squared	0.002	0.078	0.128	0.118	0.079	0.180	0.131	0.088

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.4: **STEM-degree enrollment in the analysis sample, 2000–2003 birth cohorts, definition including life sciences and medicine: gender gap and academic performance.** As Appendix Table C.3, for enrollment in a STEM degree program under the definition that also includes life sciences and medicine (Appendix Table A.1). Columns 2 to 8 exclude 4 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.

	Unconditional on post-compulsory education enrollment							
	STEM degree (math-intensive)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.0995***	−0.0980***	−0.0878***	−0.106***	−0.0943***	−0.0820***	−0.118***	−0.0808***
	(0.00235)	(0.00231)	(0.00223)	(0.00234)	(0.00229)	(0.00217)	(0.00242)	(0.00222)
6th grade Math test score			0.0537***					
			(0.000954)					
6th grade Reading test score				0.0476***				
				(0.000939)				
6th grade Math comp. adv.					0.0107***			
					(0.000971)			
10th grade Math test score						0.0787***		
						(0.00119)		
10th grade Reading test score							0.0559***	
							(0.00101)	
10th grade Math comp. adv.								0.0303***
								(0.00106)
Constant	0.147***	0.103***	0.109***	0.117***	0.101***	0.117***	0.127***	0.0952***
	(0.00265)	(0.00272)	(0.00271)	(0.00271)	(0.00271)	(0.00265)	(0.00274)	(0.00269)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	103,952	103,948	103,948	103,948	103,948	103,948	103,948	103,948
R-squared	0.028	0.071	0.100	0.093	0.072	0.130	0.101	0.078

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.5: **STEM-degree enrollment in the analysis sample, 2000–2003 birth cohorts, math-intensive definition: gender gap and academic performance.** As Appendix Table C.3, for enrollment in a STEM degree program under the math-intensive definition (Appendix Table A.1). Columns 2 to 8 exclude 4 students in singleton fixed-effect groups. Standard errors, in parentheses, are clustered by 10th-grade school. Source: Idescat administrative records; authors' calculations.